MIDTERMS: NCMB 418 - MODULE #7: NURSING CARE OF CLIENTS WITH ALTERED TISSUE PERFUSION

ANATOMY AND PHYSIOLOGY OF THE CARDIOVASCULAR SYSTEM

— consists mainly of the heart, blood, and blood vessels.
— Its major functions include:
- ➢ Transport of oxygen
- ➢ Transport of nutrients
- ➢ Removal of wastes
- ➢ Transport of hormones
- ➢ Regulation of body temperature
- ➢ Protection against infection
- ➢ Prevention of blood loss
- ➢ Maintenance of homeostasis

HEART

— cone shaped hollow muscular organ
— Located: mediastinum between the lungs
— pumps about 60ml/beat or 5L/min
— Pericardium: the protective covering of the heart

Three (3) Layers Of Cardiac Muscle Tissue:

1. Epicardium – outermost layer
2. Myocardium – middle layer
3. Endocardium – innermost layer

Four (4) Heart Chambers:

1. Right Atrium receives deoxygenated blood
2. Right Ventricle receives blood from the right atrium
3. Left Atrium receives oxygenated blood
4. Left Ventricle is the largest and most muscular chamber

Two (2) Atrioventricular Valves:

1. Tricuspid valve - right
2. Bicuspid (mitral) valve - left
   - ➢ Close at the beginning of ventricular contraction and prevent backflow
   - ➢ valves open when the ventricles relax.

Two (2) Semilunar Valves:

1. Pulmonic Semilunar Valve - bet. the right ventricle and the pulmonary artery.
2. Aortic Semilunar Valve - bet. the left ventricle and the aorta.
   - ➢ prevent blood from owing back into the ventricles during relaxation;
   - ➢ Open: contraction and Close:they begin to relax.

CORONARY BLOOD FLOW PHYSIOLOGY

ARTERY	MAIN BRANCH	PRIMARY SUPPLIES
RIGHT Coronary Artery (RCA)	Right marginal, PDA	Right ventricle, right atrium, inferior heart, often SA/AV nodes
LEFT Coronary Artery (LCA)	LAD + Circumflex (LCx)	Most of Left ventricle, interventricular septum, left atrium

— The left coronary arteries receive most of their blood flow during **DIASTOLE**.
— During systole, contraction of the LV compresses coronary vessels, especially those within the myocardium.
— During diastole, the myocardium relaxes → coronary vessels are less compressed → coronary perfusion increases.
— Therefore: ↑ Heart rate leading to ↓ diastolic filling time resulting to ↓ coronary perfusion

ELECTROPHYSIOLOGIC PROPERTIES OF THE HEART

— **AUTOMATICITY**: initiate an impulse spontaneously & repetitively
— **EXCITABILITY (DEPOLARIZATION):** respond to a stimulus
— **CONDUCTIVITY**: Transmit electrical impulses
— **CONTRACTILITY**: Contractions (systole)
— **REFRACTORINESS**: Inability to respond until repolarization

CONDUCTION SYSTEM OF THE HEART

1. SA node (60-100 times/min)
2. AV node (40-60 beats/min)
3. Bundle of His
4. R & L bundle branches
5. Purkinje fibers (20-40 beats/min)

Sequence of events during cardiac cycle:

1. Systole (contraction) emptying
2. Diastole (relaxation) filling

CARDIAC OUTPUT AND STROKE VOLUME

— **CARDIAC OUTPUT (CO)** is the volume of blood pumped by one ventricle per minute.
— **Cardiac Output = HR x SV**
— Where:
- ➢ HR: Heart rate in beats/minute.
- ➢ SV: Stroke volume in mL/beat.

— Example: HR = 75 beats/min and SV = 70 mL/beat.
75 X 70 = 5,250 mL/min
- ➢ Therefore, cardiac output is approximately 5.25 L/min.

— **STROKE VOLUME** is influenced by 3 factors:

FACTOR	DEFINITION	SIGNIFICANCE
PRELOAD	Ventricular filling and myocardial stretch at the end of diastole	Increased preload may increase stroke volume within physiologic limits; excessive volume can worsen congestion.
AFTERLOAD	Resistance against which the ventricle ejects blood	Increased systemic vascular resistance increases left ventricular workload.

CONTRACTILITY	Strength of myocardial contraction independent of preload	Reduced contractility can decrease stroke volume and cardiac output.

FUNCTIONS OF THE VASCULAR SYSTEM

— Provide conduits for blood to travel from the heart to nourish the various tissues of the body
— Carries cellular waste to the excretory organs
— Allows lymphatic flow to drain tissue fluid back into the circulation
— Returns blood to the heart for recirculation

CORONARY ARTERY DISEASE (CAD)

— Also known as coronary HEART disease (CHD)
— Describes heart disease caused by impaired coronary blood flow
— *Common cause*: atherosclerosis
— Men are more often affected than women
— Approximately 80% who die of CHD are 65+ y/o

ATHEROSCLEROSIS AND ARTERIOSCLEROSIS

— **ARTERIOSCLEROSIS**: Hardening and loss of elasticity of arterial walls.
— **ATHEROSCLEROSIS**: Fatty plaque buildup in arterial walls, narrowing the arteries.

Risk Factors for CAD

Non-modifiable Factors
— Increasing age
— Family history of premature cardiovascular disease
— Inherited predisposition

Modifiable or Treatable Factors
— Hypertension
— Dyslipidemia (LDL)
— Cigarette smoking
— Diabetes mellitus
— Physical inactivity
— Excess body weight
— Unhealthy dietary patterns
— Chronic kidney disease

Pathophysiology:

1. Endothelial dysfunction or injury: smoking, hypertension, diabetes, and abnormal lipid levels.
2. Lipid accumulation and inflammation
3. Plaque formation: Lipid-rich material and fibrous tissue
4. Progressive narrowing: limits blood flow
5. Plaque disruption and thrombosis: trigger clot formation and an acute coronary syndrome.

CAD can cause the following:

— Angina
— Myocardial Infarction (MI) = heart attack
— Cardiac dysrhythmias
— Conduction defects
— Heart failure
— Sudden death

ATHEROSCLEROSIS: ASSESSMENT

— BP (hypertension)
— Elevated cholesterol & triglycerides
— Elevated homocysteine (risk if level > 15 mmol/L)
  ➢ Blocks nitric oxide on the endothelium making cell wall less elastic & permitting plaque to build up
  ➢ Diet: B-complex vitamin rich diet (folic acid) - homocysteine
— Presence of abdominal obesity
— Elevated FBS
— Dyspnea on exertion
— Orthopnea may indicate heart failure.
— Paroxysmal nocturnal dyspnea or sudden breathlessness that awakens the patient after a period of sleep.
— Peripheral edema - venous congestion.
— Syncope - transient hypoperfusion.
— S3 in older adult may suggest heart failure or volume overload.

DIAGNOSTIC STUDIES: ELECTROCARDIOGRAM

— helps identify myocardial ischemia, acute myocardial injury, infarction patterns, and dysrhythmias.

ECG FINDING	POSSIBLE INTERPRETATION
ST-segment depression	subendocardial ischemia; correlate clinically.
T-wave inversion	occur with ischemia or other cardiac conditions.
Pathologic Q waves	myocardial necrosis from a current or previous infarction.
ST-segment elevation	acute coronary occlusion or other causes of myocardial injury; requires urgent evaluation.

CARDIAC BIOMARKERS

MARKER	SIGNIFICANCE
Myoglobin	↑ within 30 mins – 2 hrs after MI
Troponin	↑ within 3 - 6 hrs after MI & remains elevated for 21 days upon onset of attack
CK-MB	specific to myocardial tissue ↑ within 4 - 6 hrs decreases to normal within 2-3 days
LDH (LDH1)	↑ within 3-4 days & remains elevated for 14 days

Other Diagnostic Tests

1. **Exercise Stress Test**

— Assesses the heart's response to exercise
— not the first test for a suspected acute MI.

2. **Pharmacologic Stress Imaging**

— Uses medication to increase cardiac workload

3. **Doppler Ultrasonography**

— Assesses blood flow through vessels.

4. **Holter Monitoring**

— Records cardiac rhythm during daily activities, often for 24–48 hours or longer depending on the device.

5. **Echocardiography**

— Evaluates chamber size, valves, wall motion, and ventricular function.

6. **Coronary Angiography**

— A catheter is inserted through the radial artery (wrist) or femoral artery (groin).
— The catheter is advanced to the openings of the coronary arteries.
— Contrast dye is injected while X-ray images are obtained.
— Cardiologist evaluates the arteries for narrowing or blockage.
— If indicated and appropriate, PCI and stent placement may be performed

Management of Cad: Lifestyle And Risk-Factor:

— Stop smoking and avoid secondhand smoke.
— Follow an individualized heart-healthy eating pattern
— Reduce excess sodium, saturated fat, and trans fat.
— Engage in prescribed physical activity after clinical assessment.
— Control BP, blood glucose, and cholesterol.
— Maintain a healthy weight
— Medication-adherence barriers.

Surgical Procedures:

1. **Percutaneous Transluminal Coronary Angioplasty**

— to compress the plaque against the walls of the artery and dilate the vessel

2. **Laser Angioplasty**

— to vaporize the plaque

3. **Atherectomy**

— to remove the plaque from the artery

4. **Vascular Stent**

— to prevent the artery from closing and to prevent re-stenosis

5. **Coronary Artery Bypass Grafting**

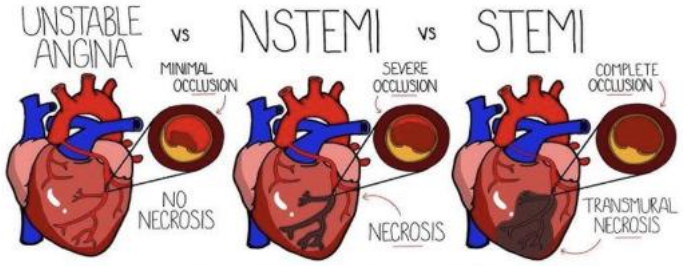
— grafting past the occluded artery to improve blood flow to the myocardial tissue at risk for

Pharmacological Management:

— **Nitrates** to dilate the coronary arteries and decrease preload and afterload
— **Calcium channel blockers** to dilate coronary arteries and reduce vasospasm
— **Cholesterol-lowering medications** to reduce the development of atherosclerotic plaques
— **Beta blockers** to control hypertension

ACUTE CORONARY SYNDROME (ACS)

— medical emergency caused by a sudden decrease in blood flow to the heart.
— Spectrum of conditions includes:



— CAUSE: blood clot blocking one of the coronary arteries.

A. ANGINA PECTORIS

— MYOCARDIAL Ischemia suppressed blood flow
— ANGINA is chest discomfort caused by transient myocardial ischemia.
  ➢ It occurs when oxygen supply is insufficient to meet myocardial oxygen demand.

TYPES OF ANGINA

Type	Characteristic	Significance
Stable angina	Predictable; relieved by rest or nitroglycerin.	Usually caused by fixed coronary narrowing.
Unstable angina	New, worsening, or occurs at rest.	Medical emergency; may progress to MI.
Vasospastic angina	Occurs at rest due to coronary spasm.	Treated with calcium channel blockers and nitrates.
Nocturnal angina	Occurs during the night.	Assess underlying ischemia.
Post-infarction angina	Recurrent angina after MI.	May indicate ongoing ischemia; urgent evaluation needed.

FUNCTIONAL CLASSIFICATION OF ANGINA

- ➢ Class I: Angina only with strenuous activity.
- ➢ Class II: Slight limitation; occurs with rapid walking, stairs, or stress.
- ➢ Class III: Marked limitation; occurs with walking 1–2 blocks or climbing one flight of stairs.
- ➢ Class IV: Angina with any activity or even at rest.

Diagnostics:

— detailed pain history
— ECG, Stress test,
— angiogram
  ➢ may be used to confirm
  ➢ describe type of angina

Management Directed Towards MI Prevention:

— Lifestyle modification (individualized regular exercise program, smoking cessation)
— Stress reduction
— Diet changes
— Avoidance of cold
— percutaneous transluminal coronary angioplasty for severe artery occlusion

Pharmacological Therapy

1. Calcium channel blockers:
— Nifedipine (Calcibloc, Adalat)
— Diltiazem (Cardizem)

2. BETA - adrenergic blockers:
— Propranolol (Inderal)
— Atenolol (Tenormin)
— Metoprolol (Lopressor)

3. Nitroglycerin (NTGs) vasodilators:
— patch (Deponit, Transderm-NTG)
— sublingual (Nitrostat
— oral (Nitroglyn)
— IV (Nitro-Bid)

4. Lipid lowering agents – statins:
— Simvastatin

5. Anti-coagulants:
— ASA (Aspirin)
— Heparin sodium
— Warfarin (Coumadin)

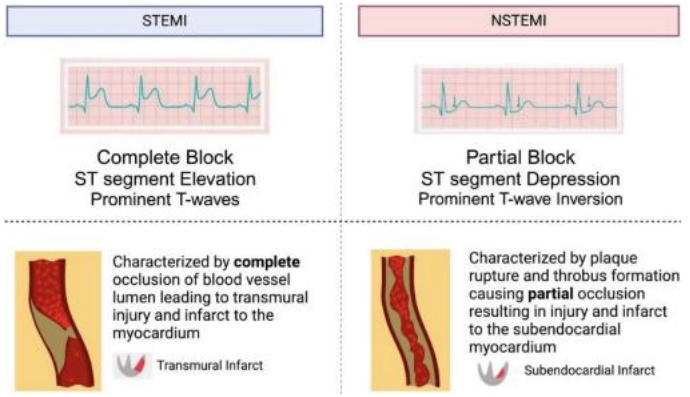
Nursing Management:
— *DIET*: low salt, low fat, low cholesterol, high fiber
— Avoid animal fats
  ➢ Recommended: White meat – chicken w/o skin, fish
— Stop smoking & avoid alcohol
— Activity restrictions (within client's limitations)
— NTGs – max of 3 DOSES AT 5 MIN INTERVALS
— Stinging sensation under the tongue for SL is normal
— Advise clients to always carry 3 tablets
— Store meds in cool, dry place, airtight amber bottles & change stocks every 6 month
— Do not discontinue the drug.
— For patches, rotate skin sites usually on chest wall
— Instruct on evaluation of effectiveness based on pain relief
— NITRATES: Inform clients that headache, dizziness, flushed face are common side effects.
— PROPRANOLOL - causes bronchospasm & hypoglycemia
  ➢ Contraindicated to asthma and DM
— HEPARIN – monitor bleeding tendencies (avoid punctures, use of soft-bristled toothbrush)
  ➢ monitor PTT levels (N: 46 - 70)
  ➢ used for 2 wks max;
  ➢ do not massage if via SC;
  ➢ have protamine sulfate available
— COUMADIN – monitor for bleeding & PT;
  ➢ always have vit K readily available
  ➢ avoid green leafy veggies

B. MYOCARDIAL INFARCTION

— Prolonged decreased blood flow to the heart results in irreversible damage to the muscle of the heart.
— The goal is to act quickly and limit the damage.

Pathophysiology:
1. Rupture or erosion of an atherosclerotic plaque followed by thrombus formation in a coronary artery.
2. Coronary blood flow decreases abruptly.
3. Myocardial oxygen supply becomes inadequate.
4. Ischemic dysfunction develops.
5. Persistent ischemia causes irreversible myocardial injury and necrosis.
6. Reduced contractility and electrical instability may lead to heart failure, dysrhythmias, or cardiogenic shock.

TYPES OF MI
1. **NSTEMI**: elevated troponin but no persistent ST-segment elevation.
2. **STEMI**: ST-segment elevation meeting ECG criteria, usually requiring urgent reperfusion.



Laboratory Test
— T–Troponin (Over 0.5 ng/mL)
— ONLY indicator of MI
— Other labs: CRP (Inflammation), CKMB, & Myoglobin

Clinical Manifestations:
— chest pain – severe crushing, constricting, "someone sitting on my chest"
  ➢ substernal radiating to left arm, neck or jaw
  ➢ prolonged (>35mins) & not relieved by rest
— Shortness of breath, profuse perspiration
— Feeling of impending doom

Pharmacologic Management:

1. Anti Clogging of Arteries
— ASA and Clopidogrel
— Antiplatelet HOLD if: Platelets 50K or LESS below 50 is risky"

2. Cholesterol Lowering Drugs
— "-Statin" Lovastatin, Atorvastatin, Simvastatin
— CAUTION: NO grapefruit
— Liver Toxic–report "clay colored stools"
— Muscle pain (Rhabdomyolysis risk) - Late sign

Management:
DURING MI–Any Chest Pain:
— Morphine
— Oxygen
— Nitro–under tongue x 3 Max
— Aspirin

AFTER MI– Clot Stabilization:
— Heparin: prevents CLOT growth (NOT dissolve)
— PTT: 46 - 70 (therapeutic range)
— Antidote: Protamine Sulfate

Pharmacological Management-Cont.:
3. CLOT BUSTER "Thrombolytics, Fibrinolytics"
— t-PA: Alteplase, Streptokinase (Allergy risk)
— Dissolves Clot ONLY (heparin does NOT)
— BLEED RISK: 8 hour window
- ➢ NO injections (IV, SQ, IM)
- ➢ NOT via central lines (CVC)
- ➢ ONLY "compressible site" (IV, PICC)

— NOT FOR:Active Bleeds: Peptic Ulcers (but menstruation is safe)
— History:
- ➢ Arteriovenous malformations
- ➢ Intracranial "Cerebral" hemorrhage
- ➢ Hypoglycemia (contraindication) INC tPA

After Myocardial Infarction– Heart Rest:
1. Beta Blockers (-lol) Atenolol
— Blocks both BP & HR (Lol = Low BP & HR)
— CAUTION:
- ➢ B–Bad for Heart Failure patients (CHF)
- ➢ B–Bradycardia (60 or Less) & BP low (HR LESS than 60)
- ➢ B–Breathing Problems "wheezing" (Asthma, COPD)
- ➢ B–Blood sugar masking "hides s/s" (Hypoglycemia)

2. Calcium Channel Blockers
— Calms BP & HR
- ➢ (AVOID Low Hr & BP)

3. Nifedipine, Diltiazem, Verapamil
— -dipine DEC BP & HR
— -zem -amil DEC Contractility and HR
— CAUTION: Citrus Fruits INC level of Drugs

4. Dilators (vasOdilators = O2 to heart)
— Nitroprusside (only for HTN crisis) & Isosorbide
— Nitro "blood flow DEC Workload"
— NO viagra Sildenafil = DEATH!
— Nitro drip: STOP if Systolic BP below 90 or 30 mmHg Drop
— SE: Headache and Syncope

Surgical Management:
1. Revascularization
— PTCA (Percutaneous Transluminal Coronary Angioplasty)
— Coronary stent implantation
— Coronary Artery Bypass Graft (CABG) – no response to medical treatment & PTCA

2. Cardiac Catheterization
— "Angioplasty, Angiogram"
— With Contrast = Kills Kidneys
- ➢ Creatinine "Kidney" (normal: 0.9 - 1.2)

— Allergy to Iodine (warm flushing normal)
— Bleeding–direct manual pressure (above site)
— NO - heparin, warfarin, ASA, clopidogrel
— STOP Metformin 48 hrs (before/after)
— REPORT:
- ➢ Creatinine Over 1.3 & Urine below 30 ml/hr
- ➢ Diminished pulses (4-12 hrs post-procedure)

— PRIORITY:
- ➢ Non palpable pedal pulse AFTER = CALL HCP
- ➢ cool leg and doppler US findings

Nursing Management:
— Promote oxygenation & tissue perfusion (place client on semi- fowler's, O2 via nasal cannula, monitor v/s changes, remind client on his activity limitations & restrictions)
— Promote comfort & rest
— Monitor the ff perimeters: v/s, ECG, rate & rhythm of pulse, effects of ADLs on cardiac status
— Diet: low salt, low cholesterol, low calories, avoid alcohol & smoking
— Take prescribe meds at regular basis
— Stress management
— Resume sexual activity after 4 - 6wks from discharge or when client can go up 2 flights of stairs without difficulty
— Assume less tiring position (non-MI partner takes active role).
— Perform sexual activity in a cool, familiar place.
— Take prescribed NTG before sexual activity
— Refrain from sexual activity after a large meal or during a tiring day.
— Moderation should be observed if palpitations, dizziness or dyspnea

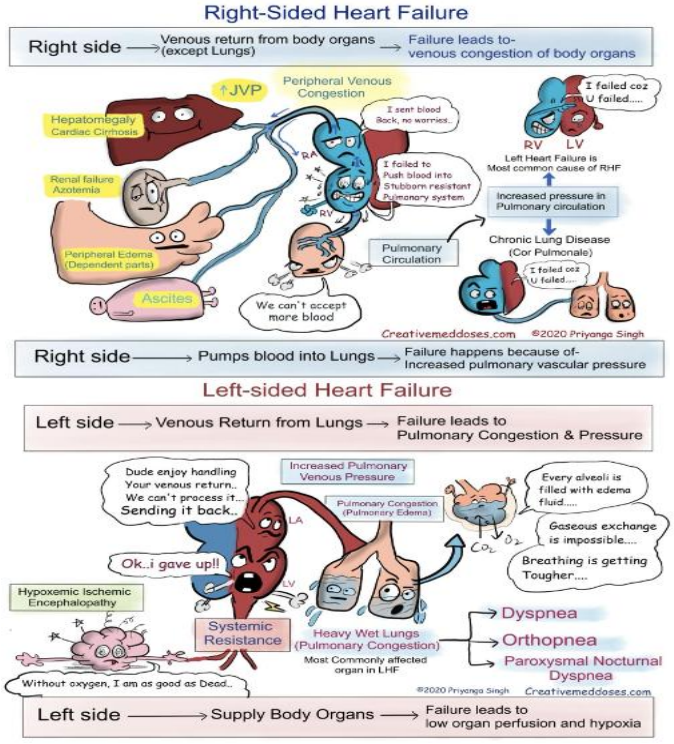
HEART FAILURE

— the inability of the heart to maintain adequate cardiac output to meet the metabolic needs of the body because of impaired pumping ability.
— failure to PUMP forward = HEAVY FLUID (lungs & body)
— Diminished cardiac output results in inadequate peripheral tissue perfusion.
— Congestion of the lungs and periphery may occur; the client can develop acute pulmonary edema.

HEART FAILURE: CLASSIFICATION
1. Acute heart failure occurs suddenly.
2. Chronic heart failure develops over time

HEART FAILURE: CLASSIFICATION (RIGHT /LEFT)

Right-Sided Heart Failure	Left-Sided Heart Failure
Dependent edema (legs and sacrum) Jugular venous distention Abdominal distention Hepatomegaly Splenomegaly Anorexia and nausea Weight gain Nocturnal diuresis Swelling of the fingers and hands Increased BP (from fluid volume excess) or decreased BP (from pump failure)	Signs of pulmonary congestion Dyspnea Tachypnea Crackles in the lungs Dry, hacking cough Paroxysmal nocturnal dyspnea Increased BP (from fluid volume excess) or decreased BP (from pump failure)



Assessment:
— Labs: BNP "Broken Ventricles"
— 300+ Mild, 600+ Moderate, 900+ SEVERE HF
— Echocardiogram
  ➢ Ejection Fraction 40% or LESS is HF! (normal-55-70%)
— LVH–Left Ventricular Hypertrophy
— Hemodynamic Monitor "Swan Ganz" (Pulmonary artery catheter)
  ➢ CVP (norm: 2-8) Over 8

Priority Management

1. Pulmonary Edema CRISIS (Lung Fluid!)
— #1 Furosemide "Body Dried" (drain fluid)
  ➢ H–HOB 45 degrees + (semi fowlers, high fowler's, orthopneic position)
  ➢ O–Oxygen
  ➢ P–Push Furosemide + Morphine, Positive inotropes
  ➢ E–End sodium & fluids (Sodium Swells)
— NO drinking fluids + STOP IV fluids

DR. BEDS
— D–Diet: Low SCC (Sodium & Cholesterol)
  ➢ Low Sodium & Fluid (2L + 2g or LESS/day)
  ➢ NO OTC meds (Cough or Flu, Antacids or NSAIDS)
  ➢ NO Canned or packaged foods (chips, sauces, meats, cheeses, wine)
— R–Risk for Falls! (Change positions slowly!)
— B–Blood Pressure & BNP (should NOT be increasing)
— E–Elevate HOB & Legs (with pillows) high fowler's
— D–Daily Weights and Is and Os (0ver 3 lbs/day or 5 lbs in 7 days) = Worsening!
— S–Stairs (No sex until able to climb 2 flights of stairs without dyspnea)
— S–Stocking (TED hose) (decreases blood pooling)
— NEVER massage calves (CHF patients)

Pharmacologic Management:
— ACTS on BP only (not HR)

1. **ACE (-pril)** Lisinopril "pril" 1st choice
2. **ARBS (-sartan)** Losartan "relax" 2nd choice
— Avoid Pregnancy
— Angioedema (Airway Risk) *only for ACE
— Cough *only Ace
— Elevated K+ (normal 3.5-5.0)

3. Calcium Channel Blockers
— Calms BP & HR (AVOID Low HR & BP) (Nifedipine)
  ➢ -dipine declined BP & HR

4. **Diuretics** Drains Fluid "Dried"
— K+ Wasting–Furosemide & Hydrochlorothiazide (caution: Low K+, Eat melons, banana & green leafy)
— K+ Sparing–Spironolactone "Spares potassium" (AVOID Salt Substitutes (inc K+), melons & green leafy)

5. **Nitroglycerin**, Isosorbide
— Caution: NO Viagra "-afil" Slidenafil = DEATH!
— Nitro drip: STOP = Systolic BP below 90 or 30 mmHg Drop
— Low BP= adverse effect (SLOW position changes)

6. **Digoxin (Inotropic)** = DEEP contraction
— Increased contractility
— Apical Pulse x 1 minute
— Toxicity (over 2.0) Vision changes, N/V
— Potassium 3.5 or less (higher r/f toxicity)

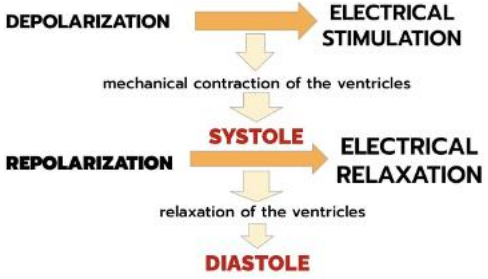
MIDTERMS: NCMB 418 - MODULE #8: NURSING CARE OF CLIENTS WITH ALTERED TISSUE PERFUSION

CARDIAC DYSRHYTHMIAS

— arrhythmias
— abnormalities in the heart's electrical activity
— It affects the heart rate, rhythm, or conduction
— potentially reducing cardiac output and tissue perfusion.

CONDUCTION SYSTEM OF HEART

1. **SA node:** Natural pacemaker; initiates
2. **AV node:** Delays the impulse, allowing the ventricles to fill.
3. **Bundle of His:** Conducts the impulse into the ventricles.
4. **Right and left bundle:** Carry the impulse down the interventricular septum.
5. **Purkinje fibers:** Distribute the impulse through the ventricles, causing contraction.



ECG WAVEFORM

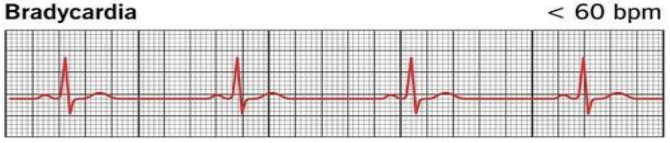
— **P wave**: Atrial depolarization
— **PR interval:** Atrial-to-ventricular conduction
— **QRS complex:** Ventricular depolarization
— **ST segment:** Period between ventricular depolarization and repolarization
— **T wave:** Ventricular repolarization
— **QT interval:** Total ventricular depolarization and repolarization

CLASSIFICATION OF COMMON CARDIAC DYSRHYTHMIAS

1. Sinus node dysrhythmias
2. Atrial dysrhythmias
3. Ventricular dysrhythmias
4. Conduction blocks

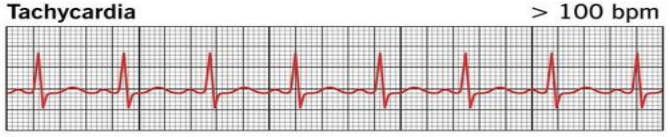
A. SINUS BRADYCARDIA

— *Rate*: Below 60 beats/min.
— *ECG*: Regular rhythm, normal P waves, each P followed by a QRS.
— *Causes*: Sleep, athletic conditioning, medications, hypothermia, or conduction disease.
— *Findings*: May be asymptomatic; severe cases can cause dizziness, hypotension, syncope, or altered consciousness.
— *Nursing Priority:* Assess perfusion and symptoms.
- ➢ Treat symptomatic bradycardia
- ➢ Atropine Sulfate



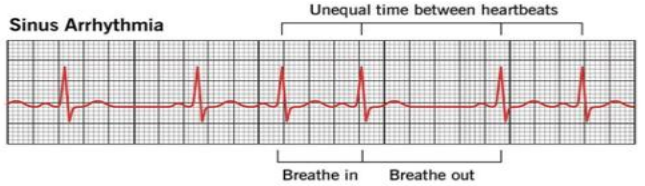
B. SINUS TACHYCARDIA

— *Rate*: Above 100 beats/min.
— *ECG*: Regular rhythm with normal P waves before each QRS.
— *Causes*: Fever, pain, anxiety, hypovolemia, anemia, hypoxia, or infection.
— *Findings*: Palpitations, dyspnea, or reduced cardiac output if sustained.
— *Nursing priority:* Identify and treat the underlying cause; do not automatically give antiarrhythmics.



C. SINUS ARRHYTHMIA

— *Rate*: Usually 60–100 beats/min.
— *ECG*: Normal P waves, but R–R intervals vary.
— *Cause*: Commonly normal variation with breathing, especially in young people.
— *Management*: Usually none is required if it is respiratory sinus arrhythmia and the patient is asymptomatic.



D. ATRIAL FIBRILLATION (AF)

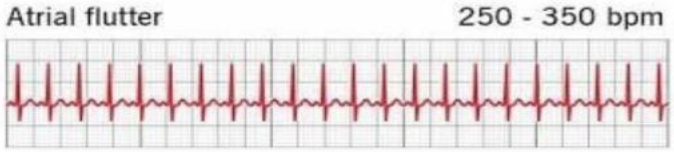
— *ECG*: No distinct P waves; irregularly irregular ventricular rhythm.
— *Causes*: Hypertension, heart failure, valvular disease, or hyperthyroidism.
— *Major risk:* Thromboembolic stroke due to atrial blood stasis.
— *Management*: Rate or rhythm control as appropriate and assessment for anticoagulation.
— *Nursing priority:* Assess hemodynamic stability, monitor the ventricular rate, and watch for stroke symptoms.



E. ATRIAL FLUTTER

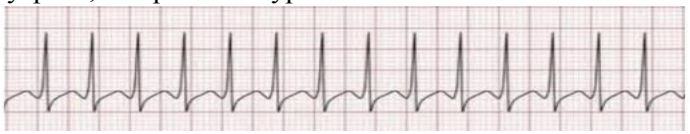
— *ECG*: Characteristic sawtooth flutter waves; ventricular rhythm may be regular or irregular.
— *Cause*: Atrial electrical reentry.
— *Risks*: Rapid ventricular response, reduced cardiac output, and thromboembolism.
— *Management*: Rate control, anticoagulation assessment, and possible cardioversion.

— *Nursing priority:* Assess pulse, blood pressure, symptoms, and ventricular response.



F. SUPRAVENTRICULAR TACHYCARDIA

— SVT is a rapid heart rhythm originating above the ventricles, commonly involving the atria or AV node.
— *Heart rate*: Usually 150–250 beats/min.
— *ECG*: Usually regular rhythm with narrow QRS complexes (less than 0.12 sec).
— *P waves:* May be hidden within the QRS complexes or difficult to identify.
— *Onset*: Often sudden, with abrupt start and stop.
— *Symptoms*: Palpitations, dizziness, chest discomfort, dyspnea, and possible hypotension.

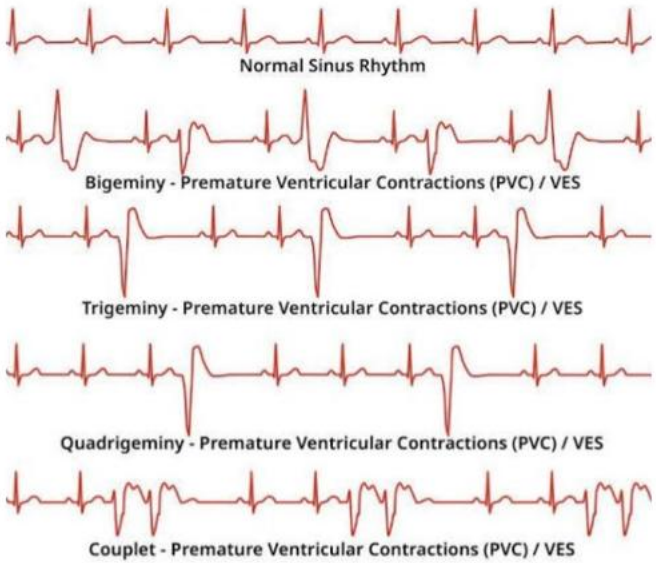


Patient Condition	Treatment
Stable SVT	Vagal maneuvers, if appropriate
SVT persists; regular, narrow -complex rhythm	IV adenosine with continuous ECG monitoring, if indicated
Unstable SVT with a pulse	Immediate synchronized cardioversion
Recurrent SVT	Beta-blockers, selected calcium channel blockers, or catheter ablation as indicated

G. PREMATURE VENTRICULAR CONTRACTIONS

— Early heartbeat originating in the ventricles
— *Cause* - increased ventricular irritability
— *ECG* - Premature, wide, bizarre-looking QRS, usually ≥ 0.12 sec
— *P wave* - Often absent before the PVC or not related to its QRS
  ➢ Often followed by a compensatory pause
— *Types*:
  1. **Unifocal** - PVCs have the same QRS shape; one ventricular focus
  2. **Multifocal** - PVCs have different QRS shapes; multiple ventricular foci
— *PVC patterns:*
  1. **Bigeminy** = every 2nd beat is a PVC
     ➢ Normal, PVC, Normal, PVC
  2. **Trigeminy** = every 3rd beat is a PVC
     ➢ Normal, Normal, PVC
  3. **Quadrigeminy** = every 4th beat is a PVC
     ➢ Normal, Normal, Normal, PVC
— *Common Triggers:*
  ➢ Hypokalemia
  ➢ Hypomagnesemia
  ➢ myocardial ischemia
  ➢ Hypoxia
  ➢ Stimulants
  ➢ Certain medications

— *Nursing Priorities*
  ➢ Assess the patient first: BP, pulse, chest pain, dizziness, and level of consciousness.
  ➢ Check ECG rhythm and PVC frequency.
  ➢ Check potassium, magnesium, oxygenation, and possible myocardial ischemia.
  ➢ Report frequent or new PVCs, especially with symptoms or underlying heart disease.
  ➢ If unstable, activate emergency response and follow the appropriate resuscitation protocol.



H. VENTRICULAR TACHYCARDIA

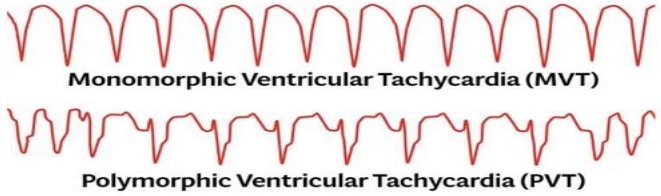
— life-threatening dysrhythmia caused by rapid electrical impulses originating in the ventricles, usually at 140–250 beats/min or faster.
— *ECG*: Usually a rapid rhythm with wide QRS complexes (≥ 0.12 seconds).
— *Occurrence*: Three or more consecutive ventricular beats constitute VT in the appropriate clinical context.
— *Major danger:* VT can deteriorate into ventricular fibrillation (VF) and cardiac arrest.
— *Types*:
  1. **Nonsustained VT:** Stops spontaneously and typically last within 30 seconds.
  2. **Sustained VT**: Lasts 30 seconds or longer and requires earlier intervention because of hemodynamic compromise.
— *Management*:
  1. **Stable VT with A Pulse:** Continuous ECG monitoring, IV access, assess reversible causes;

consider an antiarrhythmic such as procainamide, amiodarone, or sotalol as appropriate.

2. **Unstable VT with A Pulse:** Immediate synchronized cardioversion for monomorphic VT; sedation if feasible without delaying treatment.
3. **Pulseless VT:** Immediate CPR and unsynchronized defibrillation; follow the cardiac-arrest algorithm.

— *Note*:

- ➢ VT + pulse + stable → assess and prepare appropriate medication.
- ➢ VT + pulse + unstable → synchronized cardioversion.
- ➢ VT + no pulse → CPR + defibrillation.
- ➢ determine whether a pulse is present before choosing the emergency treatment.



I. TORSADES DE POINTES

— *ECG*: Polymorphic VT with QRS complexes that appear to twist around the baseline.

— *Cause*: Prolonged QT interval, often associated with medications or low potassium or magnesium.

— *Risk*: May deteriorate into VF.

— *Management*: IV magnesium for appropriate cases; immediate defibrillation in pulseless.

— *Nursing priority:* Stop and report suspected QT-prolonging causes, correct electrolytes, and prepare emergency treatment.



J. VENTRICULAR FIBRILLATION (VF)

— a life-threatening dysrhythmia in which multiple ventricular electrical impulses fire chaotically, causing the ventricles to quiver instead of contracting effectively.

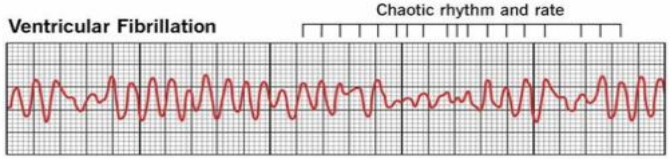
— *Result*: No effective cardiac output → no pulse → **cardiac arrest.**

— *Signs and symptoms:*

- ➢ Unconscious and unresponsive
- ➢ No definite pulse
- ➢ No effective breathing or only gasping
- ➢ No measurable blood pressure
- ➢ Chaotic ECG waveform without identifiable organized QRS complexes

— *Emergency Management of VF:*

- ➢ Start CPR: Activate emergency response; begin high-quality chest compressions and provide oxygen with bag-mask ventilation as available.
- ➢ Defibrillate immediately: unsynchronized shock
  - ○ Biphasic: 120–200 J
  - ○ Monophasic: 360 J
- ➢ Resume CPR: Continue for 2 minutes, then reassess the rhythm according to protocol.
- ➢ Give medications: Administer epinephrine and amiodarone or lidocaine as indicated; correct reversible causes.



CONDUCTION ABNORMALITIES

— PR interval is assessed for AV block.

— AV blocks - conduction through the AV nodal area is decreased or stopped.

A. FIRST-DEGREE AV BLOCK

— PR interval > 0.20 sec; every P wave conducts to a QRS



P-wave precedes each QRS-complex but interval is > 0.2 s

B. SECOND-DEGREE AV BLOCK, MOBITZ I (WENCKEBACH)

— PR interval progressively lengthens until a QRS is dropped



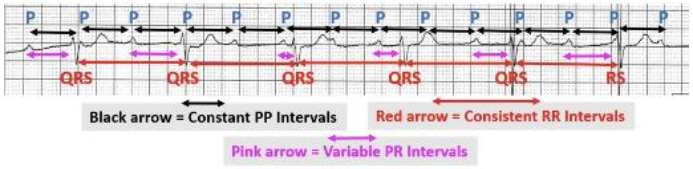
C. SECOND-DEGREE AV BLOCK, MOBITZ II

— Intermittent dropped QRS complexes without progressive PR lengthening



D. THIRD-DEGREE AV BLOCK (COMPLETE HEART BLOCK)

— P waves and QRS complexes occur independently



General Causes And Risk Factors:

Remember the common reversible causes of dysrhythmias:

— **Ischemia**: Reduced coronary blood flow or myocardial infarction.

— **Electrolyte imbalance**: Especially abnormal potassium and magnesium levels.

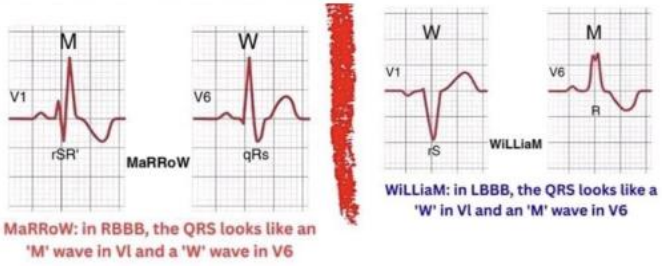
— **Hypoxia**: Inadequate oxygen supply.

— **Medications**: Digoxin, beta-blockers, calcium channel blockers, and QT-prolonging drugs.
— **Metabolic disturbances:** Acidosis, thyroid disorders, and temperature abnormalities.
— **Structural heart disease:** Heart failure, cardiomyopathy, and valvular disease.
— **Stimulants**: Excessive caffeine, nicotine, cocaine, and some sympathomimetic drugs.

Type of AV Block	Medical Management	Nursing Priority
First-degree	Usually no treatment required	Monitor ECG, symptoms, and medications
Second-degree Mobitz I (Wenckebach)	Monitor if asymptomatic; **atropine** may be considered if symptomatic	Assess BP, heart rate, and tissue perfusion
Second-degree Mobitz II	Urgent evaluation; prepare for pacing, especially if symptomatic	Continuous ECG monitoring; notify the provider promptly
Third-degree	Prepare for temporary pacing; permanent pacemaker may be required	Assess for hypotension, syncope, altered consciousness, and shock

E. BUNDLE BRANCH BLOCK (BBB)

— Delayed electrical conduction through the right or left bundle branch, causing delayed ventricular depolarization and usually a widened QRS complex.



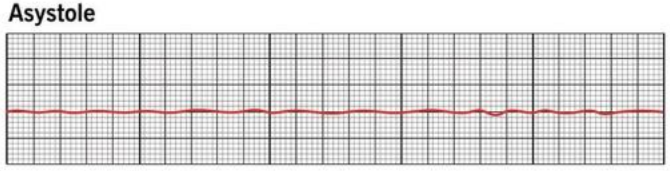
F. ASYSTOLE

— the absence of detectable ventricular electrical activity on the ECG.
— It is commonly called a flatline and is a non shockable cardiac arrest rhythm.
— Look for reversible causes of cardiac arrest.

- ➢ Hypovolemia
- ➢ Hypoxia
- ➢ Hydrogen excess
- ➢ Hypo/hyperkalemia
- ➢ Hypothermia
- ➢ Tension pneumothorax
- ➢ Cardiac Tamponade
- ➢ Toxins
- ➢ Thrombosis—coronary
- ➢ Thrombosis—pulmonary

— *Emergency Management:*

- ➢ Confirm the rhythm: Check leads, connections, and ECG gain to exclude equipment problems or fine VF.
- ➢ Check responsiveness and pulse: If pulseless, activate the emergency response and begin CPR.
- ➢ Perform high-quality CPR: Minimize interruptions and provide appropriate ventilation and oxygen.
- ➢ Administer epinephrine: 1 mg IV every 3–5 minutes according to the adult cardiac arrest algorithm.
- ➢ Identify and treat reversible causes.
- ➢ Reassess the rhythm approximately every 2 minutes; if a shockable rhythm develops, defibrillate.



ELECTRICAL CARDIOVERSION

— Delivers an electrical shock synchronized with the QRS complex to terminate certain tachyarrhythmias.
— *Indications* - Unstable tachyarrhythmias with a pulse, such as selected SVTs and monomorphic VT
— *Purpose* - Prevents shock delivery during the vulnerable T-wave period
— Preparation - ECG monitoring, IV access, sedation when feasible, airway equipment ready
— S/P - Assess rhythm, BP, consciousness, oxygenation, and skin

DEFIBRILLATION

— Delivers an unsynchronized electrical shock to terminate life-threatening ventricular dysrhythmias.
— *Indications*- Ventricular fibrillation (VF) and pulseless ventricular tachycardia (pVT)
— *Types*:

- ➢ Monophasic energy - 360 J
- ➢ Biphasic energy - Commonly 120–200 J

— *Priority* - Immediate defibrillation and high-quality CPR
— *After shock* - Immediately resume CPR
— *Safety* - Ensure everyone is clear before shock delivery

SHOCKABLE RHYTHMS

1. Ventricular fibrillation (VF)
2. Pulseless ventricular tachycardia (pVT)

— Treatment: CPR + defibrillation + ACLS medications.

NON-SHOCKABLE CARDIAC ARREST RHYTHMS

1. Asystole
2. Pulseless electrical activity (PEA)

— Treatment: CPR + epinephrine + correction of reversible causes.

VAGAL MANEUVERS

— Increase vagal stimulation, slowing conduction through the AV node and potentially terminating selected SVTs.

— *Indication*- Stable, regular SVT when appropriate

— *Effect* - Slows AV nodal conduction

— *Common method-* Valsalva maneuver

— Carotid sinus massage - Only by a trained clinician when appropriate

— *Contraindications/cautions* - Carotid bruit, known carotid disease

— *Possible complications* - Bradycardia, hypotension, syncope, and rarely serious dysrhythmias or stroke

ANTIARRHYTHMIC MEDICATIONS

- **Adenosine**: Acute, regular narrow-complex SVT
- **Atropine**: Symptomatic bradycardia
- **Magnesium sulfate**: Torsades de pointes
- **Digoxin**: Rate control in selected patients with atrial fibrillation; heart failure
- **Amiodarone**: Selected atrial and ventricular dysrhythmias
- **Lidocaine**: IV Selected ventricular dysrhythmias

MUST KNOW:

➔ Cardioversion = synchronized shock.
➔ Defibrillation = unsynchronized shock.
➔ Vagal maneuvers = slow AV nodal conduction.
➔ VF/pulseless VT = shockable.
➔ Asystole/PEA = nonshockable.

CLINICAL SITUATION	PRIORITY INTERVENTION
Stable regular SVT	Vagal maneuvers when appropriate
Persistent stable SVT	Consider adenosine when indicated
Unstable tachyarrhythmia with a pulse	Synchronized cardioversion
Ventricular fibrillation(VF)	Immediate defibrillation and CPR
Pulseless ventricular tachycardia (VT)	Immediate defibrillation and CPR
Asystole	CPR, epinephrine, and treat reversible causes; no shock
Pulseless electrical activity (PEA)	CPR, epinephrine, and treat reversible causes; no shock

MIDTERMS: NCMB 312 - MODULE #8:HEMODYNAMIC MONITORING)

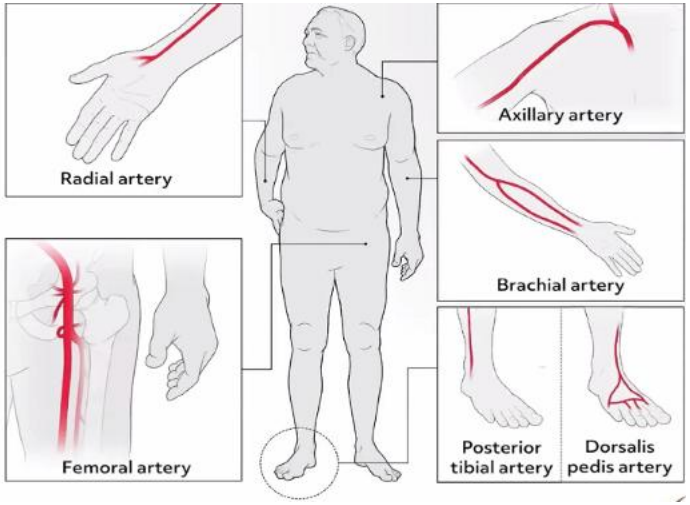
HEMODYNAMIC MONITORING

HEMODYNAMIC MONITORING

— Measure of the pressures of the cardiovascular and circulatory system

— Goals of Hemodynamic Monitoring

- ➢ Ensure adequate perfusion
- ➢ Detecting inadequate perfusion
- ➢ Titrating therapy to specific end point
- ➢ Qualifying the severity of the illness
- ➢ Differentiating system dysfunction



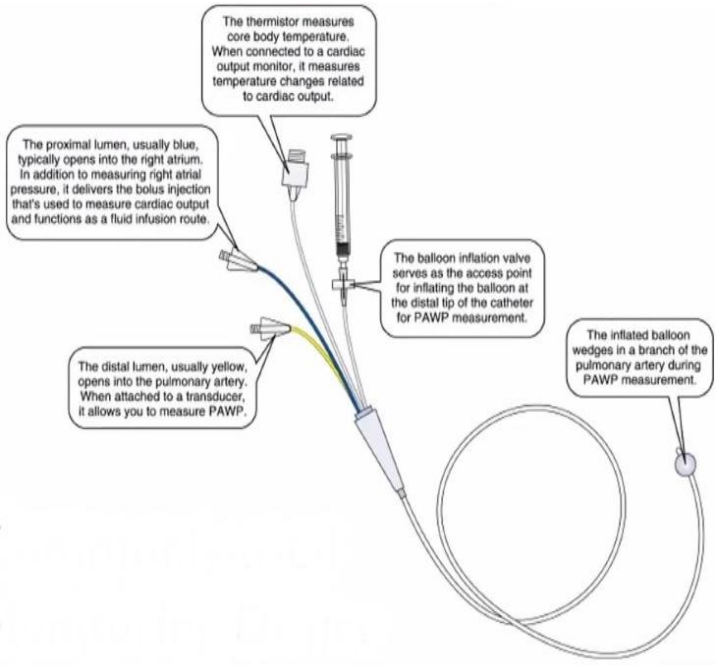
WHAT IT MONITORS

— Direct arterial BP monitoring

— CVP (Right or Left atrial pressure) monitoring

— Indirect measurements of left ventricular pressures (eg. Pulmonary artery catheters, Swan-Ganz catheters)

COMMONLY USED MONITORING DEVICE



KEY PARAMETERS

— The key parameters obtained from hemodynamic monitoring are:

- ➢ Measure of tissue oxygenation
- ➢ Cardiac output
- ➢ Stroke volume

MIXED VENOUS OXYHEMOGLOBIN

— The value of measuring SvO2 centers on the concept that:

- ➢ The amount of oxygen returning to the heart is an accurate reflection oxygenation

CARDIAC OUTPUT AND INDEX

— Cardiac Output Normal Limits: 4 - 8 L/min

— Cardiac Index Normal Limits: 2.5 - 4 L/min/m2

- ➢ < 2.2 L/min/m → threat to tissue oxygenation
  - ○ Begin treatment

$$CI = \frac{CO}{BSA} = \frac{SV \times HR}{BSA}$$

Where:

— CI = Cardiac index

— BSA = Body surface area

— SV = Stroke volume

— HR = Heart fate

— CO = Cardiac output

Note: Cl - adjustment of CO based on the size of the person

STROKE VOLUME, STROKE INDEX, AND EJECTION FRACTION

— **STROKE VOLUME**

- ➢ NV: 50 - 100 mL/beat
- ➢ Amount of blood ejected with each heartbeat

— **STROKE INDEX**

- ➢ NV: 25 - 45 mL/m2
- ➢ Measure of stroke index based on patient's size

— **EJECTION FRACTION**

- ➢ NV: >60%
- ➢ Amount of blood pumped with each contraction in relation to the amount of blood to be pumped

REGULATION OF STROKE VOLUME

Preload	Amount of stretch in the muscle just before contraction	LV - PAOP RV - CVP
Afterload	Resistance a muscle faces as it attempts to contract	LV - SVR RV - PVR
Contractility	Strength of the muscle contraction	Estimated by: SV & PAOP / CVP

INTERPRETING ARTERIAL PRESSURES

BLOOD PRESSURE

— Considered normal if the following are ruled out:
- ➢ Hypotension
- ➢ Hypertension

— Blood pressure depends on cardiac output (CO) and systemic vascular resistance (SVR)
- ➢ BP = CO x SVR

— Cardiac output depends on heart rate and stroke volume
- ➢ CO = HR x SV

— SVR = total resistance of arterioles to flow of blood
— SV =amount of blood pumped by the heart each cycle

INTERPRETING PULMONARY ARTERY PRESSURES

— Normal Limits: 25/10 mmHg
— Pulmonary HPN: > 35/20 mmHg

Pulmonary HPN
↓
fluid is forced out of the vessels
↓
interstitial & alveolar flooding
↓
interference in O2 and CO2 exchange

INTERPRETING CENTRAL VENOUS PRESSURE

CENTRAL VENOUS PRESSURE

— **Estimate of Right Ventricular End-Diastolic Pressure (RVEDP)** — is used to assess the performance of the right ventricle
- ➢ NL: 2 - 6 mmHg
- ➢ Low = hypovolemia
- ➢ Normal = normovolemia
- ➢ High = RV dysfunction

CORRELATION OF CVP AND SI

CVP	SI	Interpretation
Low	Low	Hypovolemia
Low	Normal	No hypovolemia
High	Low	RV dysfunction
High	Normal	No RV dysfunction

INTERPRETING THE PULMONARY CAPILLARY WEDGE PRESSURE (PCWP/PAOP)

— Assessment of the LV preload
— Estimates Left Ventricular End-Diastolic Pressure (LVDEP)
- ➢ NL: 8 -12 mmHg
- ➢ Low = hypovolemia
- ➢ Normal = normovolemia
- ➢ High = LV dysfunction

SYSTEMIC AND PULMONARY RESISTANCE

— Vascular resistance is frequently assumed to represent afterload
— Systemic Vascular Resistance (SVR)
- ➢ Reflects LV afterload

— Pulmonary Vascular Resistance (PVR)
- ➢ Reflects RV afterload

DERIVED PARAMETERS

SYSTEMIC VASCULAR RESISTANCE (SVR)

— Normal Limits: 900 - 1300 dynes/sec/cm^5
— Elevated:
- ➢ Primary or secondary systemic hypertension
- ➢ Compensate for low CO

— Low:
- ➢ Pathologic response to inflammation
- ➢ Hepatic disease
- ➢ Neurogenic-induced central

PULMONARY VASCULAR RESISTANCE (PVR)

— Normal Limits: 40 - 150 dynes/s/cm^5
— Elevated:
- ➢ Primary pulmonary HPN
- ➢ Secondary active pulmonary HPN
- ➢ Secondary passive pulmonary HPN

FACTORS TO CONSIDER IN HEMODYNAMIC MONITORING

1. Positioning - flat to about 40-degrees of upper-body elevation
2. Measuring CO
   - ➢ All outputs should be at 10% of each other
   - ➢ If the cardiac measurements are inconsistent, use the FLICK equation to measure CO

NURSING CARE

— Take measurements with routine intervals on, supine & at □25□HOB elevation
— When taking PAP = deflate the balloon
— When taking PCWP = inflate the balloon
— When taking CVP = supine (45□if orthopneic)
— Transducer should be at RA level (midaxillary line at 4th ICS)
— Measurements taken at point of end expiration & client should be relaxed
— Catheter connection attachments must be checked frequently that they are secure (prevent air embolism)
— Insertion site dressing must be changed as per policy
— Complications: pneumothorax, air & pulmonary emboli, fluid overload, dysrhythmia, sepsis

NORMAL VALUES

— **Cardiac Output (CO)** = 4.0 to 8.0 L/min
— **Cardiac Index (CI)** = 2.5 to 4.3 L/min/m□
— **Stroke Volume (SV)** = 50 - 100 mL/beat
— **Mean Arterial Pressure (MAP)** = 70 to 105 mmHg
- ➢ $MAP = \frac{Systolic\ BP + Two\ Diastolic\ BP}{3}$

— **Right Arterial Pressure (RAP)** = 2 to 8 mmHg

— **Pulmonary Artery Wedge Pressure (PAWP)** = 8 - 12 mmHg
— **Pulmonary Artery Diastolic (PAD)** = 10 - 15 mmHg
— **Pulmonary Vascular Resistance (PVR)** = 100 to 250 dynes/s/cm5
— **Pulmonary Vascular Resistance Index (PVRI)** - 255 to 285 dynes/s/m^2 - cm^5
— **Pulmonary Artery Mean (PAM)** = 15 to 20 mmHg
— S**ystemic Vascular Resistance (SVR)** = 800 to 1200 dynes/s/cm^5
— **Systemic Vascular Resistance Index (SVRI)** = 1970-2390 dynes/s/m - cm^5
— **Right Ventricular Stroke Work Index (RVSWI)** = 7 to 12 gm/M/M^2
— **Left Ventricular Stroke Work Index (LVSWI)** = 35 to 85 gm/M/M^2
— **Stroke Volume (SV)** = 50 to 100 mL/beat
— **Stroke Volume Index (SVI)** = 35 to 60 mL/beat/m^2
— **Right Ventricular End-diastolic Volume (RVEDV)** - 100 to 160 mL
— **Right Ventricular End-systolic Volume (RVESV)** - 50 to 100 mL
— **Right Ventricular Ejection Fraction (RVEF)** = 40% to 60%
— **Right Ventricular End-systolic Volume Index (RVESVI)** = 30 to 60 mL/m^2
— **Mixed Venous Saturation (SvO2)** = 60% - 75%
— **Oxygen Extraction Ratio (O2ER)** = 22% to 30%
— **Oxygen Extraction Index (O2EI)** = 20% to 25%

MIDTERMS: NCMB 118 - MODULE #8: CORONARY ARTERY DISEASES (ANGINA PECTORIS AND MYOCARDIAL INFARCTION)

PART 2: CORONARY ARTERY DISEASE (CAD)

PHYSIOLOGIC CHANGES OF AGING

— Aorta & arteries tend to become less distensible
— Heart becomes less responsive to catecholamines
— Maximal exercise heart rate declines
— Decreased rate of diastolic relaxation (↑ in BP is more pronounced for systolic BP than diastolic BP)
  ➢ Note that hypertension issNOTia normal age-related process
— Compensatory mechanism are delayed/insufficient = orthostatic hypotension is common
— Thickness of LV wall may increase with age due to blood vessel changes

CORONARY ARTERY DISEASE (CAD)

— Also known as coronary HEART disease (CHD)
— Describes heart disease caused by impaired coronary blood flow
— Common cause: atherosclerosis
— CAD can cause the following:
  ➢ Angina
  ➢ Myocardial Infarction (MI) = heart attack
  ➢ Cardiac dysrhythmias
  ➢ Conduction defects
  ➢ Heart failure
  ➢ Sudden death
— Men are more often affected than women
— Approximately 80% who die of CHD are 65+ y/o

Physical Assessment

— Inspection:
  ➢ Skin color
  ➢ Neck vein distention (jugular vein)
  ➢ Respiration
  ➢ Peripheral edema
— Palpation:
  ➢ Peripheral pulses
— Auscultation:
  ➢ Heart sounds (presence of S3 in adults & S4)
  ➢ Murmurs - audible vibrations of the heart & great vessels produced by turbulent blood flow
  ➢ Pericardial friction rub - extra heart sound originating from the pericardial sac
    ○ may be a sign of inflammation, infection, or infiltration
    ○ described as a short, high-pitched scratchy sound

Laboratory and Diagnostic Tests

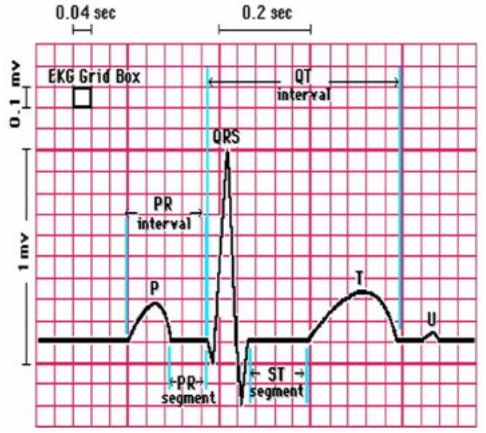
— Dyspnea
  ➢ Dyspnea on exertion - may indicate decreased cardiac reserve
  ➢ Orthopnea - a symptom of more advanced heart failure
  ➢ Paroxysmal nocturnal dyspnea - severe SOB that usually occurs 2-5hrs after onset of sleep
— Chest Pain - may be due to decreased coronary tissue perfusion or compression & irritation of nerve endings
— Edema - increased hydrostatic pressure in venous system causes shifting of plasma resulting to interstitial fluid accumulation
— Syncope - due to decreased cerebral perfusion
— Palpitations
— Fatigue

Diagnostic Tests

— ECG (Electrocardiography) - graphical recording of the heart's electrical activities; 1st diagnostic test done when cardiovascular disorder is suspected
  ➢ Waves:
    ○ P wave - atrial depolarization (contraction/stimulation
    ○ QRS complex - ventricular depolarization (changes are irreversible)
    ○ ST segment - ventricular repolarization (changes are reversible
    ○ U wave - hypokalemia
  ➢ PR interval (time for impulse to travel) = 0.12-0.20s (3-5 squares) √ for AV block
  ➢ QRS = 0.10s or (<2squares) √ for electrolyte &/or ventricular imbalance

Irregularities:

a. Absent P wave = atrial fibrillation
b. Saw-tooth pattern = atrial flutter
c. Elevated ST segment = MI
d. 3rd degree heart block = prolonged PR then progressively prolonged



CAD DIAGNOSTIC TESTS:

CARDIAC ENZYMES (CARDIAC MARKERS):

1. **Myoglobin**
   a. urine = 0 - 2mg/dL (↑ within 30mins - 2hrs after MI)
   b. blood = <70mg/dL

2. **Troponin** - regulates calcium-mediated contractile process released during MI (Troponin T & 1)
   a. blood = <0.6mg/dL - ↑ within 3-6hrs after MI & remains elevated for 21 days upon onset of attack

3. **Creatinine Kinase (CK)** - intracellular enzymes found in muscles converting ATP To ADP
   a. **CK-MB** - specific to myocardial tissue (↑ within 4-6hrs & decreases to normal within 2-3days)
      i. male = 12-70 mg/dL
      ii. female = 10-55 mg/dL

 LDH (specifically LDH,- most sensitive indicator of myocardial damage) = 45-90mg/dL - ↑ within 3-4 days & remains elevated for 14 days

STRESS TEST / TREADMILL TEST (TREADMILL STRESS TEST)

— ECG monitoring during a series of activities of patient on a treadmill

— Purposes:
- ➢ identify ischemic heart disease
- ➢ evaluate patients with chest pain
- ➢ evaluate effectiveness of therapy
- ➢ develop appropriate fitness program

— Instructions to patient:
- ➢ get adequate sleep prior to test
- ➢ avoid: caffeinated beverages, tea, alcohol, smoking, nitroglycerine on the day before until the test day
- ➢ wear comfortable, loose-fitting clothes & rubber-soled shoes on the test day
- ➢ light breakfast on the day of the test
- ➢ inform physician of any unusual sensations during the test
- ➢ rest after the test

PHARMACOLOGIC STRESS TEST

— use of intravenous injection of pharmacologic vasodilator (dipyridamole, adenosine, or dobutamine) in combination of radionuclide myocardial imaging

— To evaluate presence of significant CHD for patients contraindicated in TST

— Dipyradamole blocks cellular re-absorption of adenosine (endogenous vasodilator) & increases coronary blood flow 3-5x above baseline levels

— If with CHD, the resistance vessels distal to the stenosis already are maximally dilated to maintain normal resting flow, thus, further vasodilatation does not produce increased blood flow

— Dobutamine - used in patients with bronchospastic pulmonary disease
- ➢ increases myocardial O2 demand by increasing cardiac contractility, HR, & BP

ECHOCARDIOGRAPHY uses ultrasound to assess cardiac structure & mobility

DOPPLER U/S to detect blood flow of artery & vein specifically of lower extremities (No smoking 1hr before the test)

HOLTER MONITORING portable 24hr ECG monitoring which attempts to assess activities which precipitate dysrhythmias & its time of the day

MRI — magnetic fields & radiowaves are used to detect & define abnormalities in tissues (aorta, tumors, cardiomyopathy, pericardiac disease)

— shows actual beating & blood flow; image over 3 spatial dimensions

— Secure consent

— Assess for claustrophobia

— Remove metal items (jewelries, eyeglasses)

— Instruct client to remain still during the entire procedure

— Inform client of the duration (45-60mins)

— Contraindications: clients with pacemakers, prosthetic valves, recently implanted clips or wires

ANGINA PECTORIS / MYOCARDIAL ISCHEMIA

— **Ischemia** - suppressed blood flow

— **Angina** - to choke

— Occurs when blood supply is inadequate to meet the heart's metabolic demands

— Symptomatic paroxysmal chest pain or pressure sensation associated with transient ischemia

TYPES OF ANGINA

A. **STABLE ANGINA** - the common initial manifestation of a heart disease

— Common cause: atherosclerosis (although those with advanced atherosclerosis do not develop angina)

— Pain is precipitated by increased work demands of the heart (i.e.. physical exertion, exposure to cold, & emotional stress)

— Pain location: precordial or substernal chest area

— Pain characteristics:
- ➢ constricting, squeezing, or suffocating sensation
- ➢ Usually steady, increasing in intensity only at the onset & end of attack
- ➢ May radiate to left shoulder, arm, jaw, or other chest areas
- ➢ Duration: < 15mins
- ➢ Relieved by rest (preferably sitting or standing with support) or by use of NTG

B. VARIANT/VASOSPASTIC ANGINA (PRIN METAL ANGINA)

— 1st described by Prinzmetal & Associates in 1659

— Cause: spasm of coronary arteries (vasospasm) due to coronary artery stenosis
- ➢ Mechanism is uncertain (may be from hyperactive sympathetic responses, mishandling defects of calcium in smooth vascular muscles, reduced prostaglandin |2 production)

— Pain Characteristics: occurs during rest or with minimal exercise

- ➢ commonly follows a cyclic or regular pattern of occurrence (i.e.. Same time each day usually at early hours)

— If client is for cardiac cath, Ergonovine (nonspecific vasoconstrictor) may be administered to evoke anginal attack & demonstrate the presence & location of spasm

C. **NOCTURNAL ANGINA** - frequently occurs nocturnally (may be associated with REM stage of sleep)

D. **ANGINA DECUBITUS** - paroxysmal chest pain occurs when client sits or stands up

E. **POST-INFARCTION ANGINA** - occurs after MI when residual ischemia may cause episodes of angina

Diagnostic:
— detailed pain history
— ECG and TST
— Angiogram may be used to confirm & describe type of angina

Prevention:
— Directed towards MI prevention\
— Lifestyle modification (individualized regular exercise program, smoking cessation)
— Stress reduction
— Diet changes
— Avoidance of cold
— PTCA (percutaneous transluminal coronary angioplasty) may be indicated if with severe artery occlusion

Pharmacotherapy
— Nitroglycerin (NTGs) - vasodilators:
- ➢ Patch (Deponit,Transderm-NTG)
- ➢ Sublingual (Nitrostat)
- ➢ Oral (Nitroglyn)
- ➢ IV (Nitro-Bid)

— B-Adrenergic blockers:
- ➢ Propranolol (Inderal)
- ➢ Atenolol (Tenormin)
- ➢ Metoprolol (Lopressor)

— Calcium channel blockers:
- ➢ Nifedipine (Calcibloc, Adalat)
- ➢ Diltiazem (Cardizem)

— Lipid lowering agents - statins:
- ➢ Simvastatin

— Anti-coagulants:
- ➢ ASA (Aspirin)
- ➢ Heparin sodium
- ➢ Warfarin (Coumadin)

CLASSIFICATION OF ANGINA

— **CLASS I** - angina occurs with strenuous, rapid, or prolonged exertion at work or recreation
— **CLASS II** - angina occurs on walking or going up the stairs rapidly or after meals, walking uphill, walking more than 2 blocks on the level or going more than 1 flight of ordinary stairs at normal pace, under emotional stress, or in cold
— **CLASS III** - angina occurs on walking 1-2 blocks on the level or going 1 flight of ordinary stairs at normal pace
— **CLASS IV** - angina occurs even at rest

Nursing Care Management
— Diet instructions (low salt, low fat, low cholesterol, high fiber); avoid animal fats
- ➢ E.g. White meat - chicken w/o skin, fish

— Stop smoking & avoid alcohol
— Activity restrictions are placed within client's limitations
— NTGs - max of 3 doses at 5-min intervals
- ➢ Stinging sensation under the tongue for SL is normal
- ➢ Advise clients to always carry 3 tablets
- ➢ Store meds in cool, dry place, air-tight amber bottles & change stocks every 6 months
- ➢ Inform clients that headache,dizziness, flushed face are common side effects.
- ➢ Do not discontinue the drug.
- ➢ For patches, rotate skin sites usually on chest wall
- ➢ Instruct on evaluation of effectiveness based on pain relief

— **Propranolol** causes bronchospasm & hypoglycemia, do not administer to asthmatic & diabetic clients
— **Heparin** - monitor bleeding tendencies (avoid punctures, use of soft-bristled toothbrush); monitor PTT levels; used for 2wks max; do not massage if via SC; have Protamine Sulfate available
— **Coumadin** - monitor for bleeding & PT; always have Vit K readily available (avoid green leafy veggies)

MYOCARDIAL INFARCTION

A. UNSTABLE ANGINA/NON ST-SEGMENT ELEVATION MI

— a clinical syndrome of myocardial ischemia
— Causes: atherosclerotic plaque disruption or significant CHD, cocaine use (risk factor)
— Defining guidelines: (3 presentations)

1. Symptoms at rest (usually prolonged, i.e... >20mins)
2. New onset exertional angina (increased in severity of at least 1 class - to at least Class III) in <2months
3. Recent acceleration of angina to at least class Ill in <2months

— Diagnosis is based on pain severity & presenting symptoms, ECG findings & serum cardiac markers
— When chest pain has been unremitting for >20mins, possibility of **ST-Segment Elevation MI** is usually considered

B. ST-SEGMENT ELEVATION MI (HEART ATTACK)

— Characterized by ischemic death of myocardiäl tissue associated with atherosclerotic disease of coronary arteries

— Area of infarction is determined by the affected coronary artery & its distribution of blood flow (right coronary artery, left anterior descending artery, left circumflex artery)

— Diagnosis is based on presenting S/Sx, serum markers, & ECG (changes may not be present immediately after symptoms except dysrhythmias; PVCs/premature ventricular contractions are common after MI)

- ➢ Typical ECG changes: ST-segment elevation, Q wave prolongation, T wave inversion

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— Chest pain - severe crushing, constricting, "someone sitting on my chest"

- ➢ substernal radiating to left arm, neck or jaw
- ➢ prolonged (>35mins) & not relieved by rest

— Shortness of breath, profuse perspiration

— Feeling of impending doom

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— Death (usually within 1 hr of onset)

— Heart failure & cardiogenic shock - profound LV failure from massive resulting to low cardiac output

— Thromboemboli - leads to immobility & impaired cardiac function contributing to blood stasis in veins

— Rupture of myocardium

— Ventricular aneurysms - decreases pumping efficiency of heart & increases work of LV

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Time after Onset	Type of Injury & Gross Tissue Changes
0-0.5 hrs	Reversible Injury
1-2 hrs	Onset of irreversible injury
4-12 hrs	Beginning of coagulation necrosis
18-24 hrs	Continued necrosis; gross pallor of affected tissue
1-3 days	Total necrosis; onset of acute inflammatory process
3-7 days	Infarcted area becomes soft with a yellow-brown center and hyperemic edges
7-10 days	Minimally soft and yellow with vascularized edges; scar tissue generation begins (fibroplastic activity)
8th week	Complete scar tissue replacement

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— Initial Management: **OMEN**

- ➢ O2 therapy via nasal prongs
- ➢ Adequate analgesia (Morphine via IV - also has vasodilator property)
- ➢ ECG monitoring
- ➢ Sublingual NTG (unless contraindicated; IV may be. given to limit infarction size & most effective if given within

— Thrombolytic Therapy — best results occur if initiated within 60-90mins of onset (Streptokinase & Urokinase - promote conversion of plasminogen to plasmin)

— Anti-arrhythmics: lidocaine, atropine, propanolol

— Anticoagulants & antiplatelets: ASA, heparin

— Stool softeners

— Surgery :

1. Revascularization
   a. PTCA (Percutaneous Transluminal Coronary Angioplasty)
   b. Coronary stent implantation
   c. Coronary Artery Bypass Graft (CABG) - no response to medical treatment & PTCA
2. Resection - for aneurysm

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— Promote oxygenation & tissue perfusion (place client on semi-fowler's, O2 via nasal cannula, monitor v/s changes, remind client or his activity limitations & restrictions)

— Promote comfort & rest

— Monitor the ff perimeters: V/s, ECG, rate & rhythm of pulse, effects of ADLs on cardiac status

— Diet: low salt, low cholesterol, low calories, avoid alcohol & smoking

— Take prescribe meds at regular basis

— Stress management

— Resume sexual activity after 4-6wks from discharge or when client an go up 2 flights of stairs without difficulty

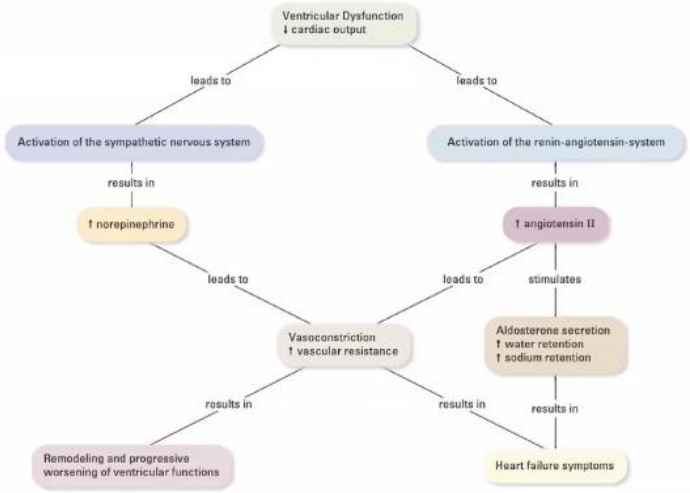
- ➢ Assume less tiring position (non-M partner takes an active role).
- ➢ Perform sexual activity in a cool, familiar place.
- ➢ Take prescribed NTG before sexual activity
- ➢ Refrain from sexual activity after a large meal or during a tiring day.
- ➢ Moderation should be observed if palpitations, dizziness or dyspnea is observed

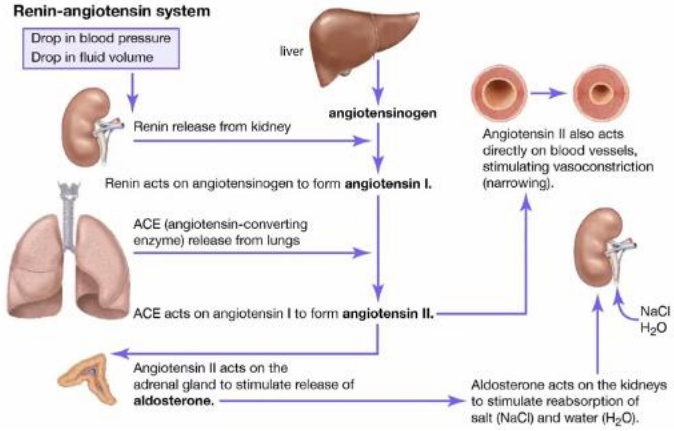
MIDTERMS: NCMB 418 - MODULE #8: DYSRHYTHMIAS

HEART FAILURE - DYSRHYTHMIA

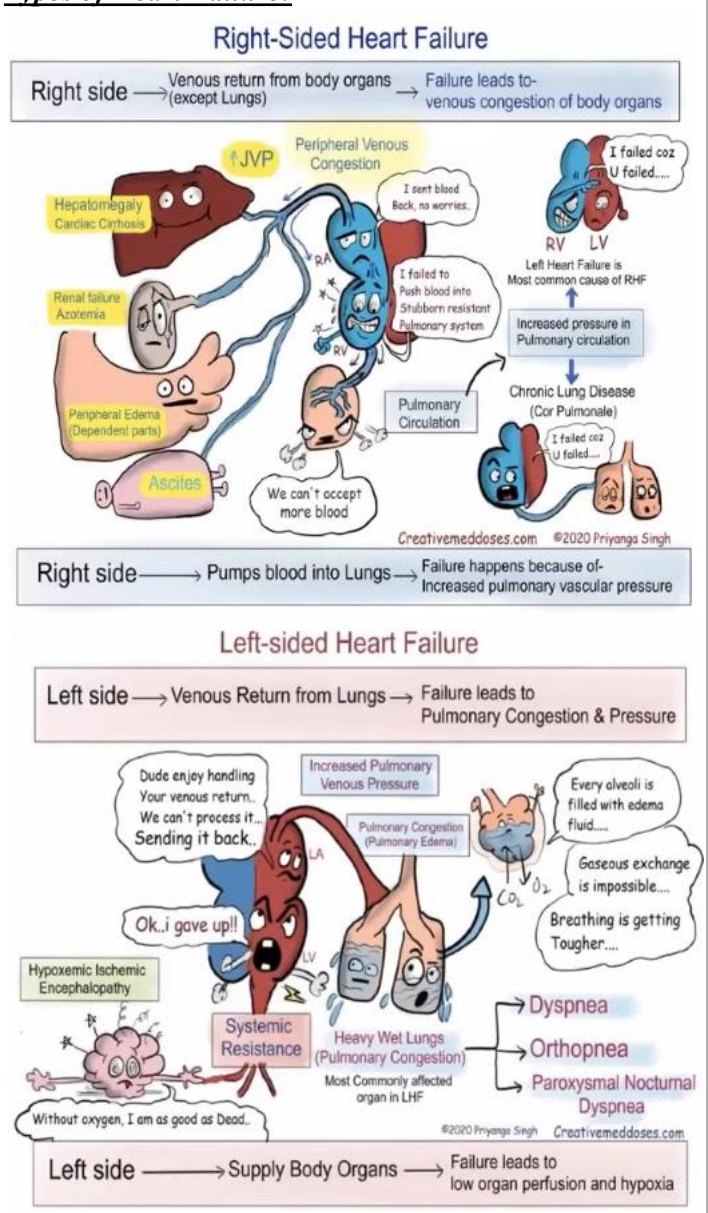
— Heart Failure - is an acute or chronic clinical syndrome that can result from any structural or functional cardiac disorder that impairs the ability of the heart to fill with or eject adequate blood throughout the body

Pathophysiology:





Types of Heart Failure:



Congestive Heart Failure Signs and Symptoms:

	Left Ventricular HF	Right Ventricular HF
General	Fatigue, weakness, lethargy, exercise intolerance	Easily fatigued
Pulmonary	Orthopnea, rhonchi, crackles, wheezes, dyspnea, frothy pink sputum, tachypnea, hacking cough worst at night, hypoxia, resp. acidosis	Dyspnea present but not as prominent
Cardiac	Tachycardia with exertion, dizziness, palpitations, irregular rhythm, s3/4 summation gallop, cardiomegaly, ↑PCWP, ↑PAP, ↑CVP, ↑/↓BP, Angina	↑PCWP, ↑PAP, ↑CVP, ↑/↓BP
Neuro	Confusion, Restlessness	
GI	Present but not as prominent	Anorexia, nausea, hepatosplenomegaly, distended abdomen, constipation
Renal	Oliguric during the day	Polyuria at night, weight gain
Extremity	Skin cool and moist, cyanosis, pallor, ↓Capillary refill, mild to moderate edema	Moderate to severe edema

Summary of Congestive Heart Failure Signs and Symptoms:

LEFT SIDE
- — **F** atigue
- — **O** thopnea
- — **R** ales/Restlessness
- — **C** yanosis/Confusion
- — **E** xtreme Weakness
- — **D** yspnea

RIGHT SIDE
- — **B** loating
- — **A** norexia
- — **C** yanosis/Cool legs
- — **O** liguria
- — **N** ausea
- — **E** dema
- — **D** istended Neck Veins

Diagnosis:

➢ **HISTORY AND PHYSICAL EXAM**
   a. Fluid status
   b. Perfusion status

c. Blood pressure
d. Weight
e. Temperature of extremities

➢ **NON-INVASIVE CARDIAC TESTING**

a. **Echocardiogram** — determines LV/RV size and function, wall motion abnormalities, valvular function and abnormalities, diastolic function, presence or absence of pericardial abnormalities or intracardiac masses, and evaluates intracardiac filling pressures.
   i. An estimate of the LVEF is also done to determine if the patient has a reduced ejection fraction (less than 40%). An echocardiogram is considered the gold standard for the diagnosis and ongoing evaluation of the patient with heart failure.

b. **Chest X-ray** — is done to assess for cardiomegaly, pulmonary congestion and may show cardiac chamber enlargement, increased pulmonary venous pressure, interstitial or alveolar edema, valvular or pericardial calcification, or coexisting thoracic diseases

c. **CBC** – Assess for anemia and infection.

d. **Serum electrolytes, BUN, Creatinine** – to assess for renal dysfunction.

e. **B-type natriuretic peptide (BNP)** – BNP and N-terminal pro-BNP levels can be used to evaluate patients with dyspnea for heart failure

f. **Liver function tests** – Elevated transaminases and LDH as well as total bilirubin levels may be present due to liver congestion.

g. **Arterial blood gas** – Alterations in oxygenation and acid-base balance may occur due to decreased perfusion and/or fluid overload.

➢ **INVASIVE CARDIAC TESTING**

a. Left sided catheterization

b. Coronary angiography

c. **Invasive Hemodynamic Monitoring** — hemodynamic values for elevated filling pressures, hypoperfusion, and altered vascular tone
   i. should only be performed in patients with severe clinical decompensation requiring escalating pressor therapy, dependence on intravenous inotropic infusions, or persistent severe symptoms despite adjustment of recommended therapies.

d. Endomyocardial Biopsy

Management:

1. Complication: Fluid Volume Excess

Interventions:
— diuretics — sodium restriction
— daily weights — intake and output

Expected Outcome:
— Normovolemia
— reduction in peripheral edema
— reduction in pulmonary edema

2. Complication: Decreased cardiac output

Interventions:
— administration of positive inotropes & vasodilators
— maintenance of sinus rhythm
— control of tachycardia

Expected Outcome: Improved cardiac output

3. **Complication: Impaired gas exchange**

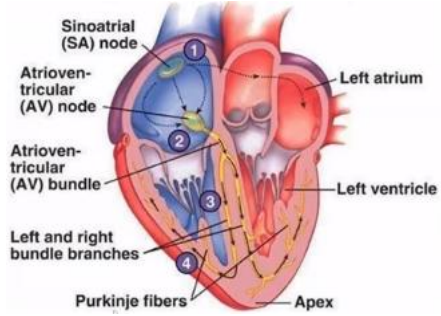
Interventions:
— high flow oxygen — CPAP or BiPAP
— mechanical ventilation — diuretics
— venodilators — positive inotropes

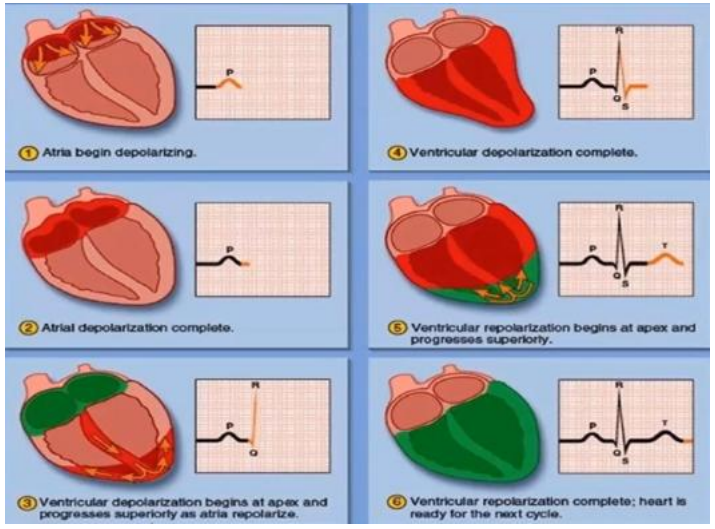
Expected Outcome: Improved gas exchange

Nursing Interventions:

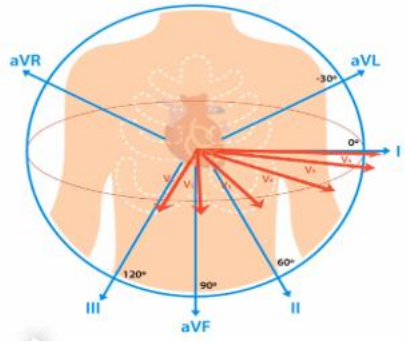
— The primary goal in managing heart failure is to maintain cardiac output.
— The Secondary goal is to decrease venous (capillary) pressure to limit edema.
— Monitor cardiac, resp., renal, & neurological status.
— Place patient in Fowler's position
— Administer O2 via cannula or mask. Intubate if necessary. Consider noninvasive positive pressure ventilation first.
— Monitor O2 saturation.
— Provide continuous ECG monitoring.
— Monitor hemodynamic status.
— Weigh daily. Assess edema. Provide sodium and fluid restrictions.
— Assess fluid and electrolyte balance.
— Provide maximum rest. Cluster activities.
— Administer diuretics to control edema and fluid retention: furosemide, bumetanide, torsemide .

DYSRHYTHMIA





TWELVE-LEAD ECG



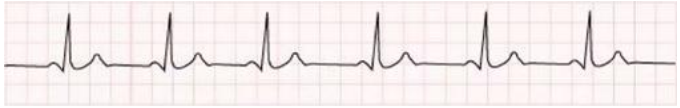
Territory	Leads
Inferior	I1, III, aVA
Lateral	I, aVL, aVR, V5, V6
Anterior	V3, V4
Septal	V1, V2

A. SINUS RHYTHM

— also referred to as normal sinus rhythm (NSR), is the rhythm that originates at the SA node.

— In summary, for NSR the ECG interpretation will be as follows:

1. Heart rhythm-regular
2. Heart rate— 60 to 100 beats per minute
3. P waves-uniform and upright
4. P to QRS ratio one to one
5. PR interval—0.12 to 0.20 second
6. QRS complex-narrow, less than 0.12 second



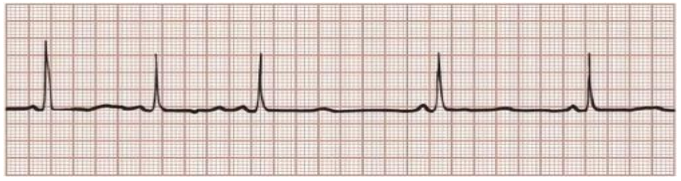
B. SINUS ARRHYTHMIA

1. Heart rhythm-irregular, heart rate increases gradually with inspiration and decreases with expiration
2. Heart rate 60 to 100 beats per minute
3. P waves-uniform and upright
4. P to QRS ratio-one to one
5. PR interval-0.12 to 0.20 second
6. QRS complex-narrow, less than 0.12 second

Etiology:

— This normal phenomenon may be a result of reflex enhancement of sympathetic tone during inspiration and parasympathetic tone with expiration.

— Sinus arrhythmia is commonly seen in healthy children but may be seen in any age group. Occasionally sinus arrhythmia may be due to nonrespiratory causes as a result of medications such as digitalis and morphine.

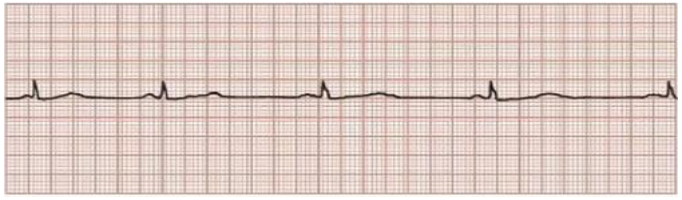


C. SINUS BRADYCARDIA

— Rhythm still originates at the SA node but at a decreased rate of less than 60 beats per minute for both the atria and ventricles.

1. Heart rhythm-regular
2. Heart rate— less than 60 beats per minute
3. P waves-uniform and upright
4. P to QRS ratio-one to one
5. PR interval—0.12 to 0.20 second
6. QRS complex-narrow, less than 0.12 second

Management: Atropine Sulfate

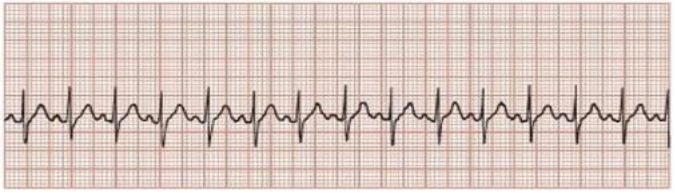


D. SINUS TACHYCARDIA

— the rhythm still originates at the SA node but at an increased rate of greater than 100 beats per minute for both the atria and ventricles.

— In summary, for ST the ECG interpretation will be as follows:

1. Heart rhythm—regular
2. Heart rate- greater than 100 beats per minute
3. P waves—uniform and upright
4. P to QRS ratio-one to one
5. PR interval—0.12 to 0.20 second
6. QRS complex-narrow, less than 0.12 second



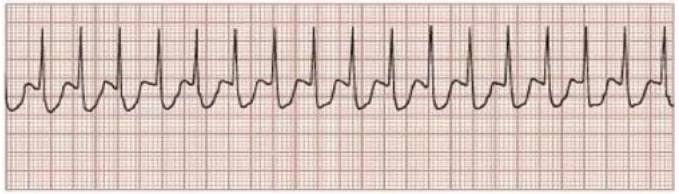
E. SUPRAVENTRICULAR TACHYCARDIA (SVT)

— is a rapid dysrhythmia that originates above the ventricles in either the atria or the AV junction.

— SVT has an atrial and ventricular rate of more than 120 beats per minute, typically 150 to 250 beats per minute.

— Rhythm is usually very regular

— No T wave



F. ATRIAL FLUTTER

1. Heart rhythm: Regular or irregular
2. Heart rate: Fast (250-350 bpm) for Atrial, but ventricular rate is often slower

3. P Wave: Not observable, but saw-toothed flutter waves are present
4. PR Interval: Not measureable
5. QRS: Normal (0.06-0.10 sec)



G. ATRIAL FIBRILLATION

1. Heart rhythm: Irregular
2. Heart rate: Very fast (> 350 bpm for Atrial, but ventricular rate may be slow, normal or fast
3. P Wave: Absent - erratic waves are present
4. PR Interval: Absent
5. QRS: Normal but may be widened if there are conduction delay



Management:
— O2 Therapy
— Cardiac monitoring
— Vagal maneuvers
— Pharmacologic therapy
- ➢ Adenosine-rapid infusion (6mg> 12mg> 12mg)
- ➢ Calcium channel blockers (Diltiazem, Verapamil); slow IV push or infusion
- ➢ Beta blockers
- ➢ Anti-arhythmics; amiodarone

— Synchronized cardioversion
— Anti-coagulant therapy

H. FIRST DEGREE ATRIOVENTRICULAR BLOCK

— a conduction delay rather than a block.
— Every impulse is conducted to the ventricles but in a delayed manner.
— In summary, for first-degree AV block the ECG interpretation will be as follows:

1. Heart rhythm-regular
2. Heart rate usually normal, but depends on the underlying rhythm
3. P waves-normal in size and shape
4. P to QRS ratio-one to one
5. PR interval-consistently prolonged from cycle to cycle, greater than 0.20 second
6. QRS complex-usually narrow, less than 0.12 second, unless there is a conduction delay



I. SECOND DEGREE ATRIOVENTRICULAR BLOCK

— In a second-degree block, not all of the impulses from the atria are conducted to the ventricles.
— The SA node is firing at regular intervals, but some of the impulses are blocked at the AV junction.
— Two types of second-degree AV blocks: **Mobitz Type I, also called Wenckebach, and Mobitz Type II**, which is more serious than type I.

A. 2ND DEGREE ATRIOVENTRICULAR BLOCK: MOBITZ TYPE I (WENCKEBACH)

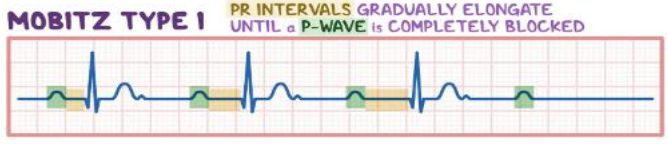
— In summary, for second-degree AV block, type I, the ECG interpretation will be as follows:

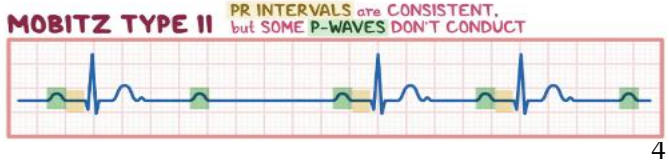
1. Heart rhythm-regularly irregular due to repeating patterns with a dropped QRS complex
2. Heart rate the atrial rate is greater than the ventricular rate due to the dropped QRS complex
3. P waves-normal in size and shape
4. P to QRS ratio-some P waves are not followed by a QRS complex
5. PR interval-lengthens with each cycle until a P wave is not followed by a QRS complex
6. QRS complex-usually narrow, less than 0.12 second

B. SECOND-DEGREE ATRIOVENTRICULAR BLOCK: MOBITZ TYPE II

— less common than type I.
— It is more serious than type I because it may suddenly progress to a third-degree AV block and because it is usually bradycardic in nature
— block occurs below the bundle of His in the bundle branches of the conduction system.
— In summary, for second-degree AV block, type II, the ECG interpretation will be as follows:

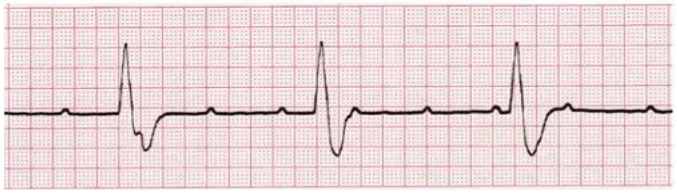
1. Heart rhythm-usually regular, but may be irregular for the ventricular rhythm
2. Heart rate—the atrial rate is greater than the ventricular rate due to the dropped QRS complexes
3. P waves-normal in size and shape
4. P to QRS ratio-some P waves are not followed by a QRS complex
5. PR interval-normal or slightly prolonged and constant for the conducted P waves
6. QRS complex-may be narrow and normal, or widened due to coexisting bundle branch block





J. THIRD-DEGREE ATRIOVENTRICULAR BLOCK (COMPLETE HEART BLOCK)

— Third-degree AV block is referred to as complete heart block because none of the impulses from the atria are conducted through the AV junction to the ventricles.
— Atria and ventricles beat independently of each other and it may also be referred to as **AV Dissociation**.



"THE HEART BLOCK POEM"
If the R is far from P. then you have a FIRST DEGREE.

Longer, longer, longer, drop! Then you have a WENCKEBACH.

If some Ps don't get through, then you have MOBITZ II.

If Ps and Qs don't agree then you have a THIRD DEGREE.

Causes of Atrio-Ventricular Block:

Causes of Temporary Block	Causes of Permanent Block
— Acute myocardial Infarction: specially **Inferior MI.** — Medications: Beta Blockers, calcium channel blockers or Digoxin . — Inflammation: myocarditis, Rheumatic fever or Lupus. — Infections: Toxoplasmosis.	— Acute myocardial infarction : specially Anterior MI. — Degeneration of Conduction system due to : advanced age or cardiac calcification of mitral or aortic valve . — Latrogenic damage: due to arrhythmia Ablation at the site of AV Junction or Valve surgery (Tricuspid valve replacement)

Clinical Manifestation:
— Usually first degree and sometimes second degree are asymptomatic.
— The most common signs and symptoms :
- ➢ Severe Bradycardia
- ➢ Hypotension
- ➢ Syncope (fainting)
- ➢ Chest pain
- ➢ Dyspnea
- ➢ Dizziness

Management and Treatment:
— General Management :
- ➢ Cardiac monitoring: for close observation
- ➢ Oxygen supply: to Manage de-saturated patients
- ➢ IV Line: To support blood pressure with fluids
- ➢ Atropine standby: to treat bradycardia specially incomplete degrees.

— Management of heart block depend on symptoms
- ➢ First degree heart block – this type usually is asymptomatic and not indicated for treatment, although they still need:
  - ○ Close observation of Hemodynamic status
  - ○ Discontinue of some medication that cause bradycardia such as :
    - ■ Beta-blockers : Concor
    - ■ Digoxins : Lanoxine
    - ■ Calcium channel blockers: Diltiazem.

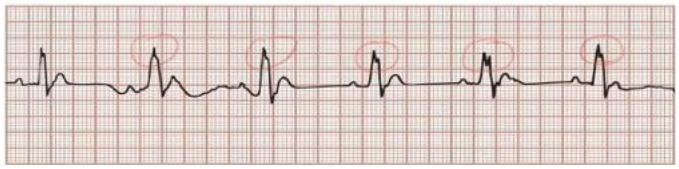
- ➢ Second Degree and Complete heart block
  - ○ Usually these degrees are associated with severe bradycardia which can be treated by atropine.
  - ○ Associated conditions should be treated correctly such as:
    - ■ Myocardial infarction.
    - ■ Electrolyte disturbance (hyperkalemia)
    - ■ Digitals intoxication.
  - ○ **Transvenous temporary pacemaker** is indicated for pt with sever bradycardia who has no effect of Atropine administration (For 24 hours : 48 hours.)
  - ○ **Transcutaneous permanent pace-maker** is indicated for chronic AV block.

Nursing Diagnosis / Priorities:
— Decrease cardiac output related to failure of the heart to pump enough blood to meet metabolic needs of the body as manifested by hypotension .
— Acute chest Pain related to decreased blood flow to myocardium through coronary arteries.
— Ineffective Tissue perfusion related to decreased cardiac output as manifested by pt syncope .
— Fatigue related to increased hypoxic tissue and slowed removal of metabolic wastes.

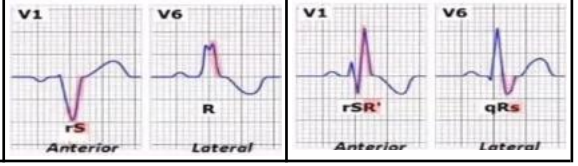
K. BUNDLE BRANCH BLOCK

— a type of intraventricular conduction block.
— BBB is a conduction delay or block in one of the two main bundle branches. Instead of both ventricles being depolarized simultaneously, the unblocked bundle branch conducts normally, then the wave of depolarization is spread from the first ventricle, cell by cell, to the second ventricle.

Etiology:
— BBB may be a temporary or permanent conduction disorder.
— A temporary BBB may occur after acute conditions such as MI or heart failure.
— A permanent BBB may occur with congenital heart disease, rheumatic heart disease, cardio-myopathy, severe aortic stenosis, or other cardiac diseases that may cause scarring of the conduction system.
— Typically, the patient with a BBB is asymptomatic. New onset of a left BBB may be indicative of acute myocardial infarction.

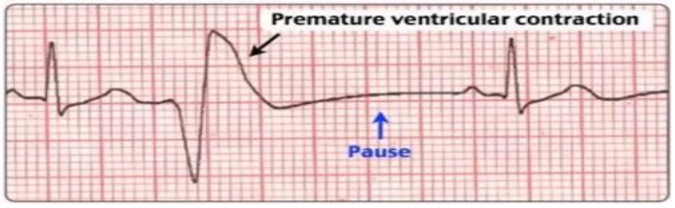


	LBBB	RBBB
Causes	Aortic stenosis, IHD, hyperkalemia, MI	RHV, PE, IHD, congenital
ECG Appearance	WILLIAM V1- W appearance V6- M appearance	MARROW V1- M appearance V6- Normal appearance, may be slurred S wave



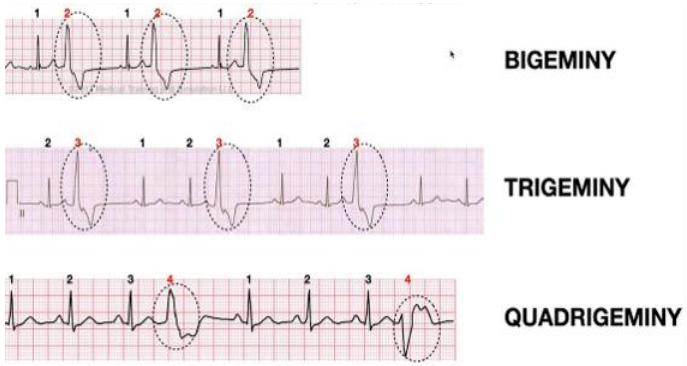
L. PREMATURE VENTRICULAR CONTRACTIONS (PVCS)

— Occur earlier than the next expected normal QRS
— Wider than a normal QRS
— QRS morphology is generally bizarre
— Preceding P wave is absent
— Deflection of the ST segment and T wave is opposite that of the QRS
— Followed by a compensatory pause



TYPES OF PVCs:

— Ventricular Bigeminy
— Ventricular Trigeminy
— Ventricular Quadrigeminy



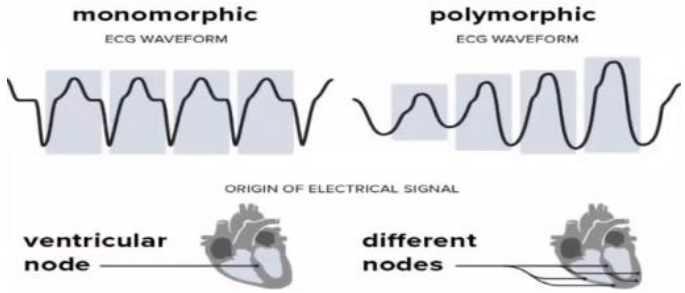
M. VENTRICULAR TACHYCARDIA

— In general, for VT the ECG interpretation will be as follows:

1. Heart rhythm-usually regular
2. Heart rate—110 to 250 beats per minute
3. P waves—usually absent
4. P to QRS ratio-the PVC does not have a P wave preceding it
5. PR interval-none
6. QRS complex- greater than 0.12 second; the shape, width, and amplitude of QRS complexes are all similar; T waves are often difficult to distinguish from QRS complexes



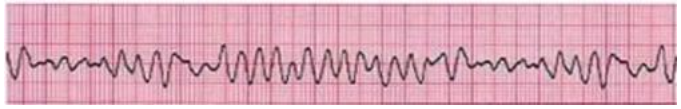
TWO TYPES OF VENTRICULAR TACHYCARDIA:



N. VENTRICULAR FIBRILLATION (VF)

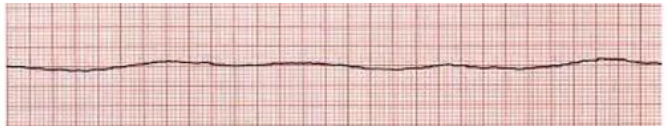
1. Heart rhythm - Irregular
2. Heart rate - 300+, disorganized
3. QRS Duration - Not recognizable
4. P Wave - Not seen

— This patient needs to be defibrillated!! QUICKLY
— Disorganized electrical signals cause the ventricles to quiver instead to contract in a rhythmic fashion
— Patient becomes unconscious as there in NO cardiac output
— This condition may occur during or after a myocardial infarct.



O. VENTRICULAR ASYSTOLE

— also referred to as **cardiac standstill**
— the absence of any electrical activity in the heart.
— There is no ventricular activity and, therefore, no QRS complexes, no ventricular contraction, and no cardiac output.

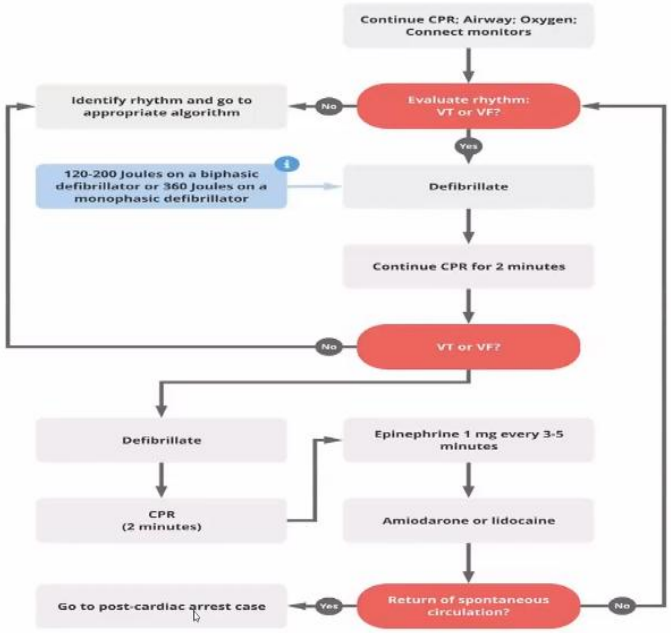


P. PULSELESS ELECTRICAL ACTIVITY (PEA)

— a clinical situation in which organized electrical activity is present, but the patient is pulseless due to absent or inadequate mechanical activity or the inability of the heart to fill or contract.
— electrical activity may produce contractions, but the contractions are too weak to produce a pulse or BP
— PEA has a poor prognosis unless a reversible underlying cause can be rapidly identified and treated.
— Note: PEA can look like any rhythm (any organized electrical activity), but if no pulse it is PEA

— When your patient is in PEA arrest don't miss the possibility of reversible causes:

H's	T's
Hypovolemia Rx: IV fluids, boluses	**Tension Pneumothorax** Rx: needle decompression
Hypoxemia Rx: O2, airway support	**Tamponade Cardiac** Rx: pericardiocentesis, thoracotomy
Hyperkalemia Rx: CaCl, Insulin + D50, NaHCO3	**Thrombosis MI** Rx: anti-coagulation + urgent PCI
Hypokalemia Rx: IV KCl 40 mEq/hr (Do not IV push)	**Thrombosis PE** Rx: Alteplase 50mg, continue CPR x 20mins
H+ Ion Excess Rx: Ventilation, bicarb for severe acidosis	**Toxins** Rx: Antidotes, supportive care, HD if applicable
Hypothermia Rx: active warming measures	



MIDTERMS: NCMB 418 - MODULE #9: NURSING CARE OF CLIENTS W/ LIFE THREATENING CONDITIONS RELATED TO GASTROINTESTINAL TRACT, LIVER, AND PANCREAS

ANATOMY AND PHYSIOLOGY OF DIGESTIVE SYSTEM

FUNCTIONS:

1. PROCESS - Mechanically and chemically
2. ABSORB - Moves nutrients, water, vitamins, and minerals
3. ELIMINATE - Removes undigested
4. SUPPORT - Provides an environment for microorganisms

KEY ORGANS AND FUNCTIONS

ORGAN	MAIN FUNCTION	
Mouth	Ingestion, chewing, saliva begins digestion	Mechanical + chemical digestion begins
Esophagus	Transports food to stomach	Uses peristalsis
Stomach	Stores and mixes food; begins protein digestion	HCl + pepsin
Small intestine	Major site of digestion and nutrient absorption	
Large intestine	Absorbs water/electrolytes and forms feces	
Rectum	Stores feces temporarily	
Anus	Eliminates feces	
Liver	Produces bile; processes absorbed nutrients; detoxification	
Gallbladder	Stores and concentrates bile	
Pancreas	Produces digestive enzymes and bicarbonate; produces insulin/glucagon	

ASSESSMENT AND DIAGNOSTICS

ASSESSMENT OF GASTROINTESTINAL TRACT METABOLISM: SUBJECTIVE DATA

ASK ABOUT		CLUES
Pain	Location, quality, severity, radiation	Epigastric → back
Nausea/ vomiting	Frequency, appearance, blood	Hematemesis
Bowel pattern	Frequency, consistency, color	Melena/ hematochezia
Appetite	Increased/decreased intake	Weight loss
Swallowing	Difficulty/pain	Dysphagia
Liver symptoms	Jaundice, itching, urine/stool changes	Hepatobiliary dysfunction
Fluid retention	Abdominal/leg swelling	Ascites/edema
Neurologic	Confusion, behavior, sleep	Hepatic encephalopathy
Bleeding	Vomiting blood/stool blood, dizziness	Hypovolemia
Alcohol	Amount, frequency, last intake	Withdrawal risk
Medication	NSAIDs, aspirin, other drugs	GI/liver injury risk
Metabolic	Thirst, urination, sweating, weakness	Glucose/electrolyte abnormalities

LABORATORY AND DIAGNOSTIC DATA

TEST	ABN. FINDING	MAY INDICATE
Ammonia	INC	Hepatic encephalopathy
Glucose	INC or DEC	Metabolic disturbance
Lipase/Amylase	INC	Pancreatic inflammation
Stool occult blood	POSITIVE	GI bleeding
AST/ALT	INC	Hepatocellular injury
ALP	INC	Cholestasis/biliary obstruction
Bilirubin	INC	Impaired bilirubin processing/excretion
Albumin	DEC	Reduced hepatic synthetic function/chronic liver disease
PT/INR	INC	Impaired clotting-factor synthesis

IMAGING STUDIES

DIAGNOSTIC	MAIN PURPOSE
Abdominal X-ray	Obstruction, perforation, abnormal gas patterns, foreign bodies
Abdominal ultrasound	Liver, gallbladder, bile ducts, pancreas; gallstones, masses, ascites
CT scan of abdomen	Tumors, inflammation, bleeding, obstruction, perforation, abscesses
FibroScan/ Elastography	Measures liver stiffness; helps assess hepatic fibrosis/cirrhosis

ENDOSCOPIC PROCEDURES

1. EGD (ESOPHAGOGASTRODUODENOSCOPY)

— Visualizes esophagus, stomach, and duodenum
— Useful for ulcers, gastritis, tumors, varices, and upper GI bleeding
— Can obtain a biopsy or control bleeding.

2. COLONOSCOPY

— Examines the colon and rectum
— Detects polyps, tumors, inflammation, and sources of lower GI bleeding
— Allows biopsy and polyp removal.

3. ERCP (ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY)

— Evaluates bile and pancreatic ducts
— Can remove bile duct stones or relieve obstruction.
— Primarily therapeutic rather than purely diagnostic.

4. LIVER BIOPSY

— Purpose: Obtain liver tissue to diagnose
— Before Procedure: Check PT/INR and platelet count; assess bleeding risk.

— Position: Supine, right arm raised.
— After Procedure: Monitor vital signs and bleeding.
— *Major complication:* Hemorrhage
— *Warning signs:* Hypotension, tachycardia, increasing abdominal/RUQ pain, pallor, falling Hgb/Hct.

SELECTED DISEASES OF GASTROINTESTINAL TRACT, LIVER AND PANCREAS

GASTROINTESTINAL BLEEDING

— a potentially life-threatening condition involving blood loss anywhere along the gastrointestinal tract.
— *Nursing concern:* Recognizing hypovolemia and impaired tissue perfusion early.

CLASSIFICATION OF GI BLEEDING

A. UPPER GI BLEEDING (UGIB)

— Bleeding originating proximal to the ligament of Treitz.

Common sites:
— Esophagus
— Stomach
— Duodenum

B. LOWER GI BLEEDING (LGIB)

— Bleeding originating distal to the ligament of Treitz.

Common sites:
— small bowel/mid-GI tract
— colon/lower GI tract.

MAJOR CAUSES OF GI BLEEDING

1. PEPTIC ULCER DISEASE

— commonly involve the:
- ➢ Stomach
- ➢ First portion of the duodenum (duodenal bulb)

Risk Factors:
— Older age
— NSAID use
— Aspirin use
— Stress

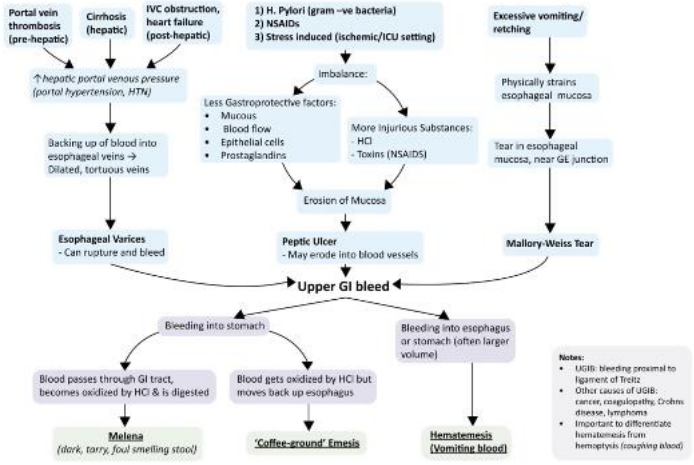
2. DIVERTICULAR DISEASE

— Diverticula are sac-like projections that develop through weak areas of the intestinal wall.
— They are commonly found in the:
- ➢ Descending colon
- ➢ Sigmoid colon
- ➢ Submucosal artery

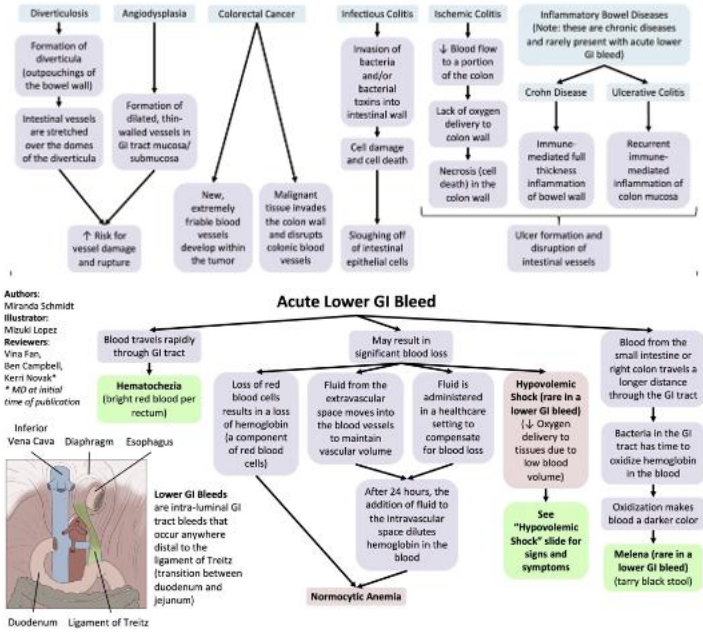
3. ESOPHAGEAL VARICES

a. Portal hypertension
b. Collateral venous dilation
c. Esophageal varices
d. possible rupture
e. Massive upper GI bleeding

PATHOPHYSIOLOGY OF UPPER GIT BLEEDING



PATHOPHYSIOLOGY OF LOWER GIT BLEEDING



CLINICAL MANIFESTATIONS: GI BLOOD LOSS

— Tachycardia
— Orthostatic hypotension
— Narrowed pulse pressure
— Decreased urine output
— Dry mucous membranes
— Flat jugular veins
— Hypotension
— ECG changes
— Poor capillary refill
— Chest pain
— Poor skin turgor
— Mental-status changes

IMPORTANT MANIFESTATIONS OF GI BLEEDING

1. Hematemesis

— Vomiting blood

2. Melena

— Black, tarry stool

3. Hematochezia

— Bright-red or maroon blood passed rectally
— lower/ upper GI bleeding

NURSING INTERVENTIONS

a. Assess ABCs immediately
   — Position side-lying if actively vomiting blood to reduce aspiration.
b. Monitor vital signs frequently
   — Watch for tachycardia, hypotension, altered LOC, and shock.
c. Establish large-bore IV access
   — Prepare for rapid administration of IV fluids and blood products as prescribed.
d. Keep patient NPO
   — Especially during active bleeding and before endoscopy/EGD.

— Monitor strict I&O - Decreased urine output
— Administer prescribed medications: PPI, Octreotide and Antibiotics
— Prepare for urgent endoscopy
— Prepare for endoscopic variceal ligation
— Avoid/flag medications that increase bleeding risk
— Monitor blood transfusion

ALCOHOL WITHDRAWAL SYNDROME

LIVER FUNCTIONS

— Metabolism (Carbohydrates, Fats, and Protein)
— Detoxification (ammonia → urea)
— Bile production
— Storage (Glycogen and vitamins)
— Immunity (Filters bacteria from blood)
— Hematologic (RBCs, WBCs, and Plts)
— Hormonal (Estrogen and Aldosterone)

ALCOHOL WITHDRAWAL SYNDROME

— occurs when a person who has been chronically consuming alcohol suddenly stops or significantly reduces alcohol intake.
— Chronic alcohol use continuously depresses the CNS. Over time, the brain compensates by increasing excitatory activity.
— When alcohol is suddenly removed: DEC Alcohol → DEC CNS inhibition + INC excitatory activity → CNS hyperactivity
— This produces symptoms such as tremors, anxiety, sweating, tachycardia, hypertension, agitation, and seizures.

TIME AFTER LAST DRINK	
6–12 hours	Tremors, anxiety, irritability, nausea, sweating, tachycardia
12–24 hours	Increasing autonomic hyperactivity; possible hallucinations
24–48 hours	Symptoms may become more severe; seizure risk
48–72+ hours	Severe cases may progress to **delirium tremens (DTs)**

CLINICAL INSTITUTE WITHDRAWAL ASSESSMENT FOR ALCOHOL, REVISED (CIWA-AR)

— tool used to assess the severity of alcohol withdrawal and help guide treatment according to the institution's protocol.
— It evaluates symptoms such as:
   ➢ Nausea/vomiting → tremor → sweating → anxiety → agitation → tactile → auditory → visual disturbances → headache → orientation

Laboratory Concerns:

— Decreased Magnesium
— Decreased blood glucose
— Decreased Potassium
— Decreased Sodium

Manifestations:

1. Hypertension & tachycardia
2. Nausea & vomiting
3. Tremors, especially hands
4. Sweating
5. Anxiety
6. Agitation
7. Tactile disturbances
8. Auditory disturbances
9. Visual disturbances
10. Headache
11. Disorientation

DELIRIUM TREMENS (DTS)

— the most severe form of alcohol withdrawal.
— It usually develops approximately 48–72 hours after cessation or substantial reduction of alcohol, although it can occur later.

Manifestations:

— Severe confusion/disorientation
— Agitation
— Severe tremors
— Tachycardia
— Fever
— Severe autonomic instability
— Hallucinations
— Profuse sweating
— Hypertension
— Possible seizures

Interventions:

— Administer medications as prescribed.
— Provide care in a nonjudgmental manner.
— Monitor vital signs and neurologic status closely; provide one-to-one supervision when indicated.
— Monitor electrolytes, glucose, and magnesium.
— Provide a quiet, low-stimulation environment.
— Allow a family member to stay when appropriate, preferably one at a time.
— Orient the client frequently.
— Explain treatments and procedures using a calm, simple approach.
— Initiate seizure precautions.
— Provide small, frequent, high-carbohydrate meals.
— Give antiemetics as prescribed.
— Monitor intake and output.
— Assist with ADLs and ambulation if stable.

MEDICATIONS FOR ALCOHOL WITHDRAWAL SYNDROME

Medication	Main purpose
Benzodiazepines	Control withdrawal + prevent seizures/DTs
Thiamine (B1)	Prevent Wernicke encephalopathy
Folate	Treat/prevent folate deficiency
Gabapentin/ Carbamazepine	Selected withdrawal cases
Haloperidol	Severe agitation/hallucinations adjunct

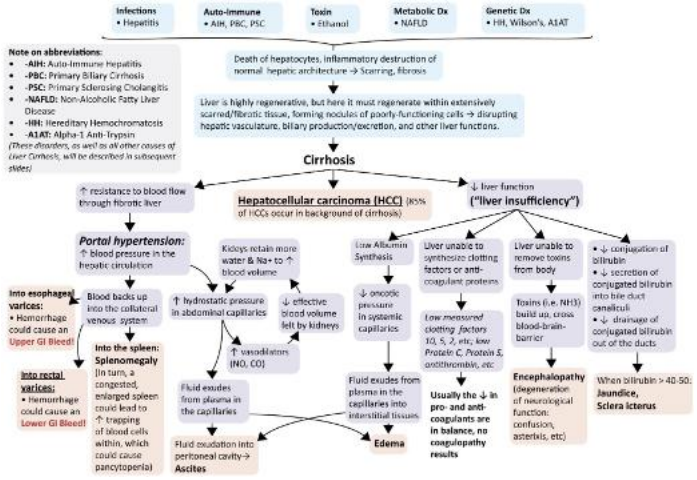
LIVER CIRRHOSIS

— chronic, irreversible scarring (fibrosis) of the liver with nodular regeneration, leading to loss of normal liver function.

Risk Factors:

— Chronic alcohol use
— Chronic viral hepatitis
  ➢ HBV, HCV
— Non-alcoholic fatty liver disease
— Biliary obstruction
— Metabolic diseases

Pathophysiology:



Diagnostic Findings:

— ↑ AST, ALT
— ↑ Bilirubin
— ↓ Albumin
— ↑ PT/INR (bleeding risk)
— Ultrasound / biopsy confirms diagnosis

Nursing Alerts:

— Confusion = hepatic encephalopathy → **priority**
— Black stools / hematemesis = variceal bleeding
— Rapid weight gain = fluid retention
— High PT/INR = bleeding risk

As cirrhosis advances, the following may occur:

— Spider nevi on the skin
— Testicular atrophy
— Impotence
— Gynecomastia
— Irregular menses
— Complications involve ruptured esophageal varices, leading to hemorrhage, circulatory shock, and acute hepatic encephalopathy.

Management:

"CIRRHOSIS CARE"

C – Check LOC
I – I&O monitoring
R – Reduce ammonia (lactulose)
R – Risk for bleeding
H – High-calorie diet
  ➢ Hepatorenal syndrome (monitor)
O – Observe complications
S – Sodium restriction
I – Infection prevention
S – Support & teaching

Medications:

— Diuretics (spironolactone, furosemide): for ascites
— Lactulose: to ↓ ammonia
— Beta-blockers → prevent variceal bleeding
— Vitamins (especially fat-soluble)

LIVER CIRRHOSIS: PARACENTESIS

— Removal of ascitic fluid from the peritoneal cavity using a needle/catheter.
  ➢ BEFORE paracentesis - have the patient void.
  ➢ DURING/AFTER - watch for hypotension.
— large-volume paracentesis - anticipate albumin as prescribed.

PANCREATITIS

PANCREAS: FUNCTION

— Acts BOTH as an digestive + hormonal organ
— Exocrine = digestion
  ➢ Enzymes + bicarbonate
— Endocrine = glucose control
  ➢ Insulin ↓ glucose
  ➢ Glucagon ↑ glucose

Pancreatic DAMAGE → digestive + diabetic problems

PANCREATITIS

— inflammation of the pancreas caused by premature activation of pancreatic enzymes, leading to self-digestion of pancreatic tissue.

Common causes:

— Gallstones (most common)
— Alcohol use
— Hypertriglyceridemia
— ERCP procedure
— Certain drugs

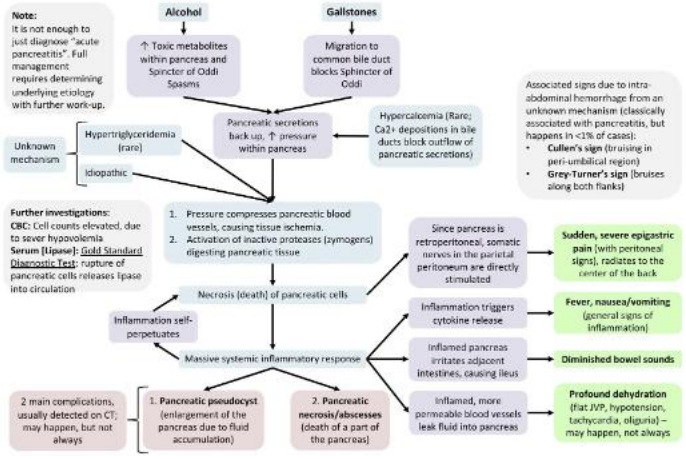
PANCREATITIS: TYPES

ACUTE PANCREATITIS = "ATTACK"
A – Abrupt
T – Temporary
T – Tender abdomen

A – ↑ Amylase/Lipase
C – Crisis
K – Kills if severe

CHRONIC PANCREATITIS= “CHANGE”
C – Continuous damage
H – Hormone loss
A – Absorption ↓
N – Nutritional deficiency
G – Gradual
E – End-stage fibrosis

Pathophysiology:



Manifestations:

A. ACUTE
— Sudden severe epigastric pain
— Nausea & vomiting
— Fever
— Tachycardia
— Hypotension (shock risk)
— *Laboratory Findings:*
  ➢ ↑ Lipase (MOST SPECIFIC)
  ➢ ↑ Amylase
  ➢ ↑ WBC
— *Severe Signs:*
  ➢ Cullen’s sign (umbilical bruising)
  ➢ Grey Turner’s sign (flank bruising)

B. CHRONIC
— Recurrent/chronic abdominal pain
— Steatorrhea
— Weight loss
— Malnutrition
— Endocrine: DM (late)
— *Laboratory findings:* Enzymes normal or slightly ↑

Medical Management:

A. ACUTE
— NPO (FIRST)
— IV fluids (aggressive)
— Pain control (opioids)
— Treat cause (e.g., gallstones)

B. CHRONIC
— Pancreatic enzymes
— Low-fat diet
— Small frequent meals
— Stop alcohol
— Insulin if needed

Pharmacologic Management:

— IV fluids - Lactated Ringer’s (LR)
— Opioid analgesics - Fentanyl or Morphine
— Antiemetics - Ondansetron
— Electrolyte replacement - Replace K^+, Ca^{2+}, Mg^{2+} as indicated.
— Insulin - during significant hyperglycemia
— Pancreatic enzyme replacement - Not routine

MIDTERMS: NCMB 418 - MODULE #10: NURSING CARE OF CLIENTS W/ LIFE THREATENING CONDITIONS RELATED TO METABOLIC EMERGENCIES

DIABETES MELLITUS (DM)

PANCREAS

— endocrine portion consists mainly of the islets of Langerhans.

DIABETES MELLITUS (DM)

— a chronic metabolic disorder characterized by hyperglycemia resulting from problems with insulin secretion, insulin action, or both.

1. Insulin DEC or insulin resistance INC
2. Glucose cannot effectively enter cells
3. Blood glucose INC
4. Cellular glucose utilization is impaired
5. Complications develop over time

Type 1	Autoimmune β-cell destruction → absolute insulin deficiency	Requires insulin
Type 2	Insulin resistance + progressive β-cell dysfunction	Most common type

Signs And Symptoms:

Classic 3 Ps:

1. Polyuria — excessive urination
2. Polydipsia — excessive thirst
3. Polyphagia — excessive hunger

Other manifestations:

— Fatigue
— Weight loss
— Recurrent infections
— Dehydration
— Blurred vision
— Poor wound healing
— Pruritus

Diagnostic Tests:

Several tests may be employed:

1. Fasting Plasma glucose (FPG) or; Fasting Blood Sugar (FBS)
2. Oral Glucose Tolerance Test (OGTT)
3. Random Plasma Glucose Test or; Random Blood Sugar (RBS)
4. Haemoglobin A1C (HbA1C)
5. Capillary blood glucose (CBG)

Criteria For The Diagnosis Of Diabetes Mellitus:

— Fasting plasma glucose ≥ 126 mg/dL
— Oral glucose tolerance test (OGTT): 2-hour plasma glucose ≥ 200 mg/dL
— Haemoglobin A1C 6.5% or higher

Treatment:

ORAL AND INJECTABLE DRUGS - Standards of Care recommends a four-step approach:

STEP 1: At diagnosis, initiate lifestyle changes plus metformin.
STEP 2: continue lifestyle changes plus metformin, and add a second drug.
STEP 3: Progress from step 2 to a three-drug combination (inclusive of METFORMIN).
STEP 4: If 3 drugs including basal insulin fails to achieve treatment goals after approximately 3 months - combination injectable regimen inclusive of insulin and possibly a GLP-1 receptor agonist.

DIABETIC KETOACIDOSIS (DKA)

— an acute, life-threatening complication of diabetes characterized by:

1. Hyperglycemia
2. Ketosis
3. Metabolic acidosis

Precipitating Factors:

NO INSULIN + STRESS

— New-onset diabetes
— Missed or inadequate insulin
— Infection
— Myocardial infarction
— Stroke
— Trauma
— Surgery
— Severe emotional/physiologic stress
— Insulin pump failure
— Certain medications, particularly glucocorticoids

Pathophysiology:

1. Insulin deficiency
2. Glucose cannot adequately enter cells
3. Hyperglycemia
4. Osmotic diuresis
5. Polyuria
6. Dehydration + hypovolemia

At the same time:

1. Insulin deficiency + increased counter-regulatory hormones
2. Fat breakdown (lipolysis)
3. Free fatty acids → ketone bodies
4. Ketosis
5. Metabolic acidosis
6. Kussmaul respirations + fruity/acetone breath

Laboratory Findings:

Blood glucose	Usually >250 mg/dL
Ketones	Present
pH	<7.30
HCO_3^-	<18 mEq/L
Osmolality	Increased
Dehydration	Significant
Onset	Usually rapid

DKA Electrolyte Problem — POTASSIUM

— Despite total-body potassium depletion, the patient's serum K^+ may initially be normal or high.

1. Insulin deficiency + acidosis causes:
→ K^+ shifts out of cells
→ serum K^+ may increase

BUT:

2. Hyperglycemia causes osmotic diuresis, resulting in:
→ urinary K^+ loss
→ total-body K^+ depletion

Clinical Manifestation:

Early / common

— Polyuria
— Polydipsia
— Polyphagia
— Weakness
— Fatigue
— Dehydration
— Weight loss
— Nausea and vomiting
— Abdominal pain

Progressive/severe DKA

— Tachycardia
— Hypotension
— Dry mucous membranes
— Altered mental status
— Kussmaul respirations
— Fruity/acetone breath
— Severe dehydration
— Coma

Treatment Priorities

1. IV Fluids - 0.9% sodium chloride
2. Insulin - Regular Insulin
   - ➢ > 250 IV insulin
   - ➢ < 200 SQ + D50/50 IV
3. Potassium/Electrolyte management
4. Identify and treat the cause

Treatment For K+ Imbalance:

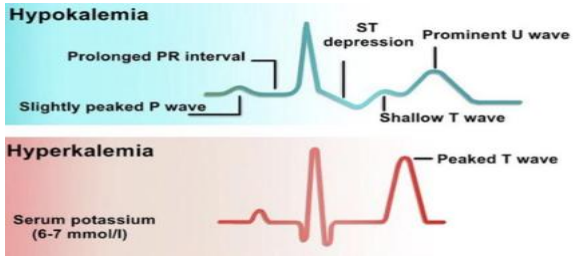
— #1 Monitor the heart
— Never PUSH
— 10 - 20 mg MAX per HR
— Slow INFUSION

Hypokalemia

— Flattened T wave
— ST depression
— Prominent U wave

Hyperkalemia

— Peaked T wave
— ST elevation



HYPEROSMOLAR HYPERGLYCEMIC STATE

— characterized primarily by:

1. Severe hyperglycemia
2. Hyperosmolality
3. Profound dehydration

— Significant ketoacidosis is usually absent or much less prominent.

Pathophysiology:

— enough insulin to suppress substantial ketone formation but not enough insulin to prevent severe hyperglycemia.

1. Relative insulin deficiency
2. Severe hyperglycemia
3. Osmotic diuresis
4. Massive water loss
5. Severe dehydration
6. Increased serum osmolality
7. Neurologic manifestations

Laboratory Findings:

Blood glucose	Usually ≥600 mg/dL
Ketones	Minimal/absent
pH	Usually >7.30
HCO_3^-	Usually >18 mEq/L
Serum osmolality	>320 mOsm/kg
Dehydration	Very severe
Neurologic changes	Common
Onset	Days to weeks

Medical Management:

Main priorities

1. Aggressive fluid replacement
2. IV insulin
3. Electrolyte replacement/monitoring
4. Treat precipitating cause

Fluid resuscitation is a major priority, monitor carefully for:

— Fluid overload
— Heart failure
— Pulmonary edema
— Changes in neurologic status

SYNDROME OF INAPPROPRIATE ANTIDIURETIC HORMONE

ANTIDIURETIC HORMONE

— Pituitary = master gland
— Posterior pituitary = storage
— ADH = controls water balance
— ADH is made in hypothalamus
— ADH acts on kidneys
— SIADH = low sodium
— DI = high urine output

SYNDROME OF INAPPROPRIATE ANTIDIURETIC HORMONE

— Excessive ADH secretion from the pituitary gland even in subnormal serum osmolality.

1. cannot excrete dilute urine
2. retain fluids
3. dilutional hyponatremia.

Causes:

— Bronchogenic carcinoma
— Severe pneumonia
— Pneumothorax
— Head injury
— Brain surgery or tumor
— Infection
— Some medications

Clinical Manifestation:
— STOPs urination (LOW urine output)
— STICKY & THICK "urine"
— HIGH Sp. Gravity 1.030+
— SOAKED Inside "Low & Liquidy" Labs
— HYPO osmolality (LOW)
— HYPOnatremia below 135 Na+ (LOW)
— SODIUM Low!! (Headache Early Sign)
— SEIZURES- Headache & Confusion
— SEVERE HIGH blood pressure

Medical Management:
— Eliminating the underlying cause

Pharmacological Management:
— **Diuretics**- Furosemide (Lasix)
— **Demeclocycline**- drugs that render the kidneys less sensitive to ADH
— STOP ALL FLUIDS + GIVE Salt + Diuretics
➢ (NO IV or drinking) + (IV 3% Saline + Eat Salt)

Nursing Management:
— Monitor fluid intake and output, daily weight, urine and blood chemistries, and neurologic status
— Provide supportive measures and explanations of procedures and treatments to assist patients to deal with this disorder
— Restrict fluid intake as indicated
— Regularly assess mental status

MIDTERMS: NCMB 418 - MODULE #10: NURSING CARE OF CLIENTS W/ LIFE THREATENING CONDITIONS RELATED TO RENAL SYSTEM

FUNCTION OF THE RENAL SYSTEM

— **Urine formation**: filtration, reabsorption, secretion, excretion
— **Excretion of waste products**: urea, creatinine
— **Regulation of electrolytes:** sodium, potassium, calcium, phosphate, magnesium
— **Regulation of acid–base balance:** reclaims bicarbonate, excretes hydrogen
— **Control of water balance:** ADH
— **Control of blood pressure:** RAAS
— Regulation of red blood cell production
— Synthesis of vitamin D
— Secretion of prostaglandins
— **Regulates calcium and phosphorus balance:** tied to vitamin D activation and phosphate excretion

Organ	Main Function	
Kidneys	Filter blood and produce urine	
Ureters	Transport urine to the bladder	
Urinary Bladder	Store urine	
Urethra	Eliminate urine from the body	
Nephrons	Microscopic functional units of the kidneys.	Filter blood, reabsorb needed substances, secrete waste, and form urine.
Glomerulus	Network of capillaries inside the renal corpuscle.	Filters water and small solutes from. the blood.
Bowman's Capsule	Cup-shaped structure surrounding the glomerulus.	Collects the fluid filtered from the blood
Renal Tubules	Tubular structures of the nephron.	Reabsorb water and useful substances and secrete selected waste products.
Collecting Ducts	Tubes that receive fluid from multiple nephrons.	Adjust the final concentration of urine and transport it toward the renal papillae.
Renal Artery	Blood vessel that supplies the kidney.	Brings blood to the kidney for filtration.
Renal Vein	Blood vessel that drains the kidney.	Returns filtered blood to the inferior vena cava

ACUTE KIDNEY INJURY (AKI) OR ACUTE RENAL FAILURE (ARF)

— AKI/ARF - a rapid loss of renal function due to damage to the kidneys.
— Depending on the duration and severity - potentially life-threatening metabolic complications (metabolic acidosis/ f/e imbalances)
— Usually reversible. If untreated, it may lead to permanent kidney damage.
— *Most common cause* - ischemia and nephrotoxins.
— Possibly caused by iatrogenic causes such as nephrotoxic medications, radiologic contrast dye, and surgical shock.

Conditions that Reduce Blood Flow to the Kidney and Impair Kidney Function:
— Hypovolemia;
— Hypotension;
— reduced cardiac output and heart failure;
— obstruction of the kidney or lower urinary tract by tumor, blood clot, or kidney stone;
— bilateral obstruction of the renal arteries or veins.

NUMEROUS CAUSES OF ACUTE RENAL FAILURE:

1. Acute bilateral kidney disease, such as glomerulonephritis, which reduces GFR.
2. Severe and prolonged circulatory shock or heart failure, which results in tubule necrosis.
3. Nephrotoxins such as drugs, chemicals, or toxins, which cause tubule necrosis and obstruction of blood flow.
4. With burns, the damaged erythrocytes break down in the circulation, releasing free hemoglobin that may accumulate in the tubules, causing obstruction.
5. Hemoglobin also is toxic to tubule epithelium, causing inflammation and necrosis
6. When skeletal muscle is crushed in an accident, myoglobin is released with similar effects.

COMMON DRUGS TO STOP IN AKI

"DAAMN"
- ➢ Diuretics
- ➢ ACEi/ARBS (pril/sartan)
- ➢ Aminoglycosides (mycins)
- ➢ Metformin
- ➢ Nsaids

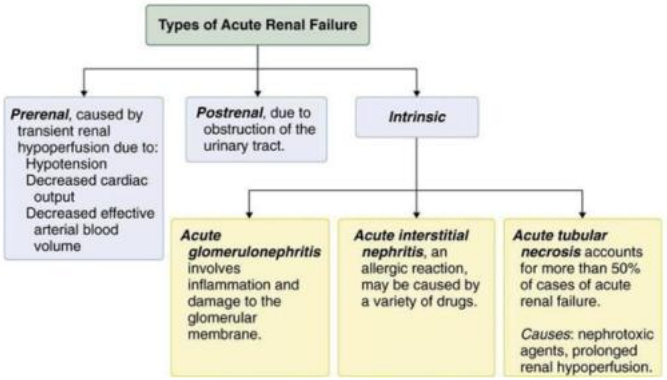
Note: ACEi/ARBs - Nephroprotective in CKD & Nephrotoxic in AKI

CAUSES OF ACUTE RENAL FAILURE

PreRenal
— Hypoperfusion of kidney
— Volume depletion
— Impaired cardiac efficiency
— Vasodilation

IntraRenal
— Actual damage to kidney tissue
— Prolonged renal ischemia
— Nephrotoxic agents
— Infectious process

PostRenal
— Obstruction to urine flow
— Urinary tract obstruction
— Calculi (stones), tumors
— Benign prostatic hyperplasia
— Blood clots



PHASES OF ACUTE KIDNEY INJURY

1. ONSET
— Benign with initial insult and ends when oliguria develops
— Initial phase of injury 1-3 days

2. OLIGURIC PHASE
— accompanied by an increase in the serum concentration of substances usually excreted by the kidney
— Urine output <400 cc/24 hrs for 3 days – 2 weeks
— BUN, creatinine
— Edema, hypertension
— Hyperkalemia, Hyponatremia, Hyperphosphatemia
— Metabolic Acidosis

3. DIURETIC PHASE (LAST- 1 WEEK)
— marked by a gradual increase in urine output, which signals that glomerular filtration has started to recover.
— Urine output 3 – 5 L/day for 10 days
— Elevated BUN & creatinine, Elevated BP, Hypokalemia

4. RECOVERY PHASE
— signals the improvement of renal function and may take 3 to 12 months
— Avoid nephrotoxic drugs
— May lead to CRF

Laboratory Profile Of ARF:
— Elevated BUN and creatinine
— Sodium retention but may be deceptive due to water retention
— Potassium increased
— Phosphorus increased
— Calcium decreased
— H&H decreased
— Sp. Gravity decreased and fixed

Sign and Symptoms:
— Irritability
— Headache
— Anorexia
— Tingling of extremities
— Lethargy that can progress to stupor & coma
— Sudden dramatic drop in urinary output
— Restlessness, twitching, convulsions
— Skin pallor, anemia & increased bleeding time
— Ammonia odor breath & perspiration
— Generalized edema
— Hypertension which can progress to pulmonary edema & CHF

Medical Management:
— Hyperkalemia—closely monitor electrolytes
➢ Kayexalate/Sorbitol
— IV dextrose (D50/50), insulin and calcium may help shift K+
— Cautious administration of any medication that can be nephrotoxic
— Monitor ABGs and acid-base balance - Metabolic acidosis
➢ Sodium Bicarbonate
— Monitor phosphate levels
— Calcium carbonate to control hyperphosphatemia
— Antacids or H2 receptor antagonist to prevent GI ulcer - Ranitidine
— Renal-dose dopamine to improve renal perfusion.
— Fluid overload - Furosemide or Mannitol

Nutritional Therapy:
— Caloric requirements are met with high-carbohydrate meals, because carbohydrates have a protein-sparing effect
— Foods and fluids containing potassium or phosphorus (e.g., bananas, citrus fruits and juices, coffee) are restricted.
— Following the diuretic phase, the patient is placed on a high-protein, high-calorie diet and is encouraged to resume activities gradually.

Nursing Management:
— Diet: protein restriction
— High carbohydrates & restricted potassium
— Total parenteral nutrition
— Monitor I & O
— Observe for oliguria followed by polyuria
— Weight patient daily & observe for edema
— Monitor electrolyte imbalance
— acidosis & hyperkalemia

CHRONIC KIDNEY DISEASE (CKD)

— Progressive, usually irreversible deterioration of kidney function lasting at least 3 months.
— *No. 1 cause*– Diabetes mellitus
— Accumulation of metabolic wastes in the blood due to severe kidney dysfunction.

Pathophysiology:

1. EARLY STAGES OF DECREASED RESERVE (60% of nephrons lost)

— decrease in GFR
— serum creatinine - consistently higher
— serum urea levels that are normal
— remaining nephrons appear to adapt
  ➢ increasing their capacity for filtration.

2. SECOND STAGE (around 75% of nephrons lost), or that of renal insufficiency - change in blood chemistry

— GFR is decreased to approximately 20% of normal
  ➢ significant retention of nitrogen wastes
— Tubule function is decreased
  ➢ failure to concentrate the urine, control the secretion and exchange of acids and electrolytes.
— Osmotic diuresis occurs
  ➢ excretion of large volumes of dilute urine (Hyposthenuria)
— Erythropoiesis is decreased
  ➢ cardiovascular system compensate for these effects

3. UREMIA, OR END-STAGE RENAL FAILURE (more than 90% of nephrons lost), occurs when GFR is negligible.

— Fluid, electrolytes, and wastes are retained
— marked oliguria or anuria develops.
— Regular dialysis or a kidney transplant is required

KEY SIGNS IN ADVANCED CKD

— Oliguria: Decreased urine output.
— Hyperkalemia: Elevated serum potassium; risk of fatal dysrhythmias.
— Fluid overload: Edema, hypertension, and pulmonary edema.
— Uremia: Nausea, confusion, seizures, or coma
—Metabolic acidosis:Reduced kidney excretion of acids.

GFR CATEGORIES/ STAGE OF CKD

1. GFR greater than or equal to 90mL/min/1.73 m2 -
   ➢ Kidney damage w/normal or increased GFR
2. GFR = 60-89 - mild decrease in GFR
3. GFR = 30-59 - moderate decrease in GFR
4. GFR = 15-29 - severe decrease in GFR
5. GFR < 15 - Kidney failure

CLINICAL MANIFESTATIONS PER SYSTEM

Every body system is affected

➢ Cardiovascular - hypertension (RAAS), heart failure, pulmonary edema, pericarditis, MI
➢ Pulmonary - crackles, Kussmaul, pleuritic pain\
➢ Dermatologic - severe pruritus, uremic frost (urea crystals)
➢ Gastrointestinal - n/v, anorexia, uremic fetor (ammonia breath), constipation or diarrhea
➢ Neurologic - LOC changes, confusion, seizures, agitation, neuropathies
➢ Hematologic - anemia, thrombocytopenia
➢ Musculoskeletal - muscle cramps, renal osteodystrophy, bone pain, bone fractures
➢ Metabolic changes - urea and creatinine, sodium, potassium, acid-base, calcium and phosphorus

Early Signs Of Chronic Renal Failure:

— Increased urinary output (polyuria), manifested as frequency and nocturia
— General signs: anorexia, nausea, anemia, fatigue, unintended weight loss, and exercise intolerance
— Bone marrow depression and impaired cell function caused by increased wastes and altered blood chemistry
— High blood pressure - RAAS

As the kidneys fail completely (end-stage failure), uremic signs appear:

— Oliguria
— Dry, pruritic, and hyperpigmented skin, easy bruising
— Peripheral neuropathy - abnormal sensations in the lower limbs
— Impotence and decreased libido in men, menstrual irregularities
in women
— Encephalopathy (lethargy, memory lapses, seizures, tremors) > Ammonia/Urea
— congestive heart failure, arrhythmias
— failure of the kidney to activate vitamin D
— hypocalcemia and hyperphosphatemia with osteodystrophy
— Osteoporosis and tetany
— Possibly uremic frost on the skin and a urine-like breath odor in the terminal stage or if infection is present

Diagnostic Tests:

KEY indicators of chronic renal failure:

— Metabolic acidosis becomes decompensated (serum pH below 7.35). (ABG)
— Azotemia refers to the presence of urea or other excess nitrogen wastes in the blood
— elevated serum creatinine and urea levels. (BUN/CREA)
— Anemia becomes severe. (CBC)
— Serum electrolyte levels may vary: (serum ELEC.)
  ➢ hyponatremia and hyperkalemia
  ➢ hypocalcemia and hyperphosphatemia.

Medical Managements:

— Phosphorus binders - Renvela (Pro)
  ➢ Generic name: sevelamer
— Antihypertensives
— Antiseizure—valium or dilantin
— Erythropoietin - Epoetin alfa
— Iron supplements, Folic acid, B12
— Diet - > carbohydrates and fats, vitamins, restrict protein X

Nutrition Therapy:
— Regulate protein intake
— Allow only protein with high biological value (eggs & meat)
— Fluid intake to balance fluid losses
  ➢ Fluid allowance (500-600Ml) more than the previous days 24 hour urine output
— Some restriction of potassium
— Adequate caloric intake and vitamin supplement

Maintain Fluid & Electrolyte Balance:
— Monitor VS - I&O
— Weight the patient daily
— Assess presence & extent of edema
— Auscultate breath sounds
— Administer phosphate – binding agent Calcium carbonate or acetate

Prevent Infection & Injury:
— Meticulous skin care
— Encourage activity but avoid fatigue
— Protect from exposure to infectious agent
— Avoid aspirin
— Soft toothbrush

DIALYSIS
— to prevent complications, such as hyperkalemia, metabolic acidosis, and pulmonary edema.
— allows for liberalization of fluid, protein, and sodium intake
1. Hemodialysis circulates the patient's blood through an artificial kidney [dialyzer] to remove waste products and excess fluid
2. Peritoneal dialysis uses the patient's peritoneal membrane as the semipermeable membrane to exchange fluid and solutes. peritoneal cavity is filled up with hypertonic solution to PULL solutes out

— Before Procedure:
  1. Take Weight
  2. Warm solution
— Infection Risk: Sterile technique PRIORITY
— Peritonitis: Key Signs to REPORT
  ➢ Fever
  ➢ Tachycardia
  ➢ Cloudy drainage
— Respiratory Distress: Key Signs
  ➢ Crackles in lung bases
  ➢ Rapid respirations
  ➢ Dyspnea
    ○ Priority Intervention- First action: Raise HOB
— Insufficient outflow
  ➢ Assess Patient = abdomen: distention & Constipation
  ➢ Assess Device = catheter kinks & obstructions
  ➢ Intervention: Reposition to side-lying position

HEMODIALYSIS
— Machine version of the kidney
— Helps to clean the blood by filtering waste & flushes out excess fluid and electrolytes

Medication to hold during Hemodialysis:
— Antihypertensives
  ➢ Ace & Arbs: Lisinopril, Losartan
  ➢ Beta Blockers: Atenolol
  ➢ Ca Channel Blockers: Nifedipine, Verapamil, Diltiazem
  ➢ Diuretics: Furosemide, Hydrochlorothiazide
  ➢ Dilators: Nitroglycerine
— Washed Out:
  ➢ Antibiotics
  ➢ Digoxin
  ➢ Water-soluble vitamins (B, C, and folic acid)

Deadly Complication:
DIALYSIS DISEQUILIBRIUM SYNDROME (DDS)
— Restless & disoriented
— Vomiting
— Headache
— *Priority Action:*
  ➢ Stop or slow infusion
  ➢ Report to Provider

Nursing Interventions:
— Care for Fistula (AV shunt)
  ➢ Squeeze or Grip: "Rubber ball" "sponge"
  ➢ Pitting edema = Normal
  ➢ Not Normal:
    ○ Pale skin "pallor"
    ○ Paresthesia "Numbness or tingling"
    ○ Pulses diminished
    ○ Poor cap refill
    ○ Pain (distal to shunt)
— Monitor for Infection, Bleeding, and feel for a thrill
— NO restrictive clothing or jewelry (watch)
— NO BP on affected arm
— NO sleeping on arm
— NO creams or lotions
— NO lifting over 5 Ibs